



STANFORD UNIVERSITY SHOP SAFETY PROGRAM

ENVIRONMENTAL HEALTH AND SAFETY
480 Oak Road
Stanford CA 94305

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Shop Safety Program Stanford University

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1. PURPOSE / INTRODUCTION

Besides Land, Buildings and Real Estate (LBRE) and Residential & Dining Enterprises (R&DE) shops, there are more than 40 Academic, Teaching and Research-Support Shops and Makerspaces on Stanford University campus and satellite / field locations. Areas of shop / makerspace activity include metal-working and machining, wood-working and carpentry, electrical / electronic work, glass blowing / forming, scenery and set building, art and architecture building / modeling, research / device building, mechanics and vehicle building / maintenance, 3D plastic and metal printing, among others.

This program integrates safe work practices required by Cal-OSHA and the University's Injury and Illness Prevention Program (IIPP) into the operations of Departments that own and/or manage Academic Shops and Makerspaces at Stanford University. Under the direction of the University President's Office via the University Committee on Health and Safety (UCHS), as well as the Environmental Health and Safety (EH&S) Department, this program provides shop-management tools and guidance, outlines safe work requirements and procedures, and guides training and recordkeeping for all work conducted in Academic Shops / Makerspaces on property owned and/or leased by Stanford University regardless of personnel's affiliation with the University. All employees, students, visitors and contractors working in Academic Shops / Makerspaces must be familiar with and follow the requirements of this program.

Integration and implementation of this program is accomplished through use of the Attachments and forms at the end of this program, ongoing work of the Shop Safety Committee, as well as any additional safety management activities a Shop Manager may deem are needed to ensure shop safety in their area of oversight.

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2. APPLICABILITY & SCOPE

With the exception of LBRE and R&DE who have their own Shop Safety Management Programs, this Program applies to Academic Shops and Makerspaces, defined as a physical space or lab that is leased, rented, owned and/or managed by a Department where one or more of the following operations are routinely conducted by Stanford Faculty, Staff, Students or Contractors:

- **Metal working** including sheet metal forming, machining, grinding, riveting, cutting, threading, casting, forging, heat-treating, quenching, welding, brazing, soldering, 3D Printing, sputtering, etc...
- **Carpentry and woodworking** including cutting, drilling, sanding, carving, routing, grinding, planing, gluing, bonding, fastening, etc...
- **Surface modification and coating** including sandblasting, painting, surface preparation, laminating, burning, etching, masking, etc....
- **Glass work** including glass blowing, glazing, annealing, tempering, bonding, grinding, drilling, hot-work with glass materials, etc...
- **Electrical / electronic** work including equipment building, circuit design / building, wiring, control system building/repair, soldering, etc.



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- **Plastics and Fabrics** work including machining, bending, burning, bonding, extruding, cutting, drilling, gluing, melting, forming, sewing, stitching, 3D Printing, etc...
- **Concrete / Rebar** work including form-building, rebar cutting, concrete mixing / pouring, destructive testing, etc...
- **Equipment Development / Model Building** work including model building, hydraulics building / use, compressed air use, equipment building, research equipment repairs / maintenance, etc...

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ShLab – Exemption from the Shop Safety Program

Many labs / student-use spaces have lower hazard shop / makerspace processes (e.g. only hand tool, small hand-held power tools, table-top 3-D plastic printer, etc.) that are ancillary to primary work activities of the space. For such locations, full enrollment in the Shop Safety Program is voluntary. Owner/managers of such spaces must still designate a “safety coordinator” for the Shop Safety Committee (SSC) to communicate with, but may opt-out of sending a representative from the work area to SSC meetings. Opt-out and contact information are documented using [Attachment 3](#).

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3. DEFINITIONS

Authorized Person / Authorized User – Used interchangeably throughout this document. Defined as a person who, by reason of their need to work in the shop combined with their experience and knowledge of shop safe work practices, is “Authorized” to use shop equipment / processes and physically be present / work in shop areas defined as “Restricted Areas”. This process is conducted for each “Authorized User” by the Shop User’s Supervisor / Instructor combined with the Shop Manager (if a different person), and documented by the Shop Manager using Attachment 1.

Makerspace – Used interchangeably throughout this document with the term “shop”. Defined as a space designed for prototyping, designing, building, creating, assembling, crafting or other kinds of physical activities that results in producing a physical object. Typical activities in Makerspaces may include (but aren’t limited to) sewing, stitching, gluing, laminating, painting, 3-D Printing, circuit-board making, soldering, etc. All Makerspaces that have any of the processes noted in the “Applicability / Scope” section of this program fall under the purview of this Shop Safety Program.

Owner Department – Any department, excepting LBRE and R&DE, that occupies and oversees space utilized for the purpose of shop work activities.

Restricted Area – An area of a shop that has controlled access and in which only “Authorized Persons” are allowed to freely work without supervision or escort. All of a shop’s high-hazard processes must be located within a Restricted Area.

“ShLab” – A location within a research lab where shop equipment, tools and processes are used as part of lab research. Usually these processes are lower-hazard and may fall under the “ShLab – Exemption” from the Shop Safety Program as noted in the “Applicability / Scope” section of this program.



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Shop – Used interchangeably with “Makerspace” throughout this document. Defined as a location where any of the activities or operations described in the “Applicability / Scope” section of this program occur. For the safety and risk control purposes of this program, Makerspaces are included under the “Shop” designation.

Student Organization – A student-run organization that is focused on a subject, project or objective, and advised / guided by a faculty member. In the context of this program, a Student Organization that uses or builds its own Shop / Makerspace would be designated by the Faculty Advisor completing a Student Organization MOU (Attachment 3) outlining the details of the Student Organization shop.

Visitor - Any person who is not an “Authorized Person” is known as a “Visitor” in a shop. A visitor may not enter or remain in “Restricted Areas” of any shop without the escort of an “Authorized Person”.

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4. ABBREVIATIONS

EH&S	– Environmental Health and Safety Department
LBRE	– Land, Buildings and Real Estate Department
R&DE	– Residential and Dining Enterprises Department
UCHS	- University Committee on Health and Safety
SO	- Student Organization
SSC	– Shop Safety Committee

5. ROLES AND RESPONSIBILITIES

Faculty, Staff and Students

All Faculty, Staff and Students, regardless of title and position within the University, who work in a shop / makerspace must be “Authorized” by a Responsible Person in the shop’s Owner Department to work in the space before they use shop equipment, conduct hazardous work or enter a “Restricted Area” of any shop. Under no circumstances is anyone allowed to conduct work in a shop without prior “Authorization”. Any person who enters a shop but has not been through the qualification and training process within a specific shop by the Shop Manager to make them an “Authorized Person”, is called a “Visitor”.

Authorized Person

1. Any person who works with shop equipment and processes must be “authorized” by their PI/Instructor/Supervisor, as well as the “Shop Manager” prior to working unescorted in a shop. This is a documented process using [Attachment 1](#).



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2. Must be trained on, and remain current to, each Shop's specific safe-work requirements including but not limited to general shop rules, personal protective equipment use, and solo work. Training on these Stanford University Shop Safety Requirements are documented using [Attachment 2](#).
3. May only work on or with shop equipment and conduct shop processes that they have been 'Authorized' to do.
4. May enter and work unescorted in a shop's "Restricted Area".

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Shop / Makerspace Manager

This person may be referred to as "Shop / Makerspace Manager" or "Shop Manager" interchangeably throughout this program document. They assure requirements of this program are integrated into the operations of their Shop / Makerspace by implementing and following the "Shop Manager Guideline" in Attachment 8 of this program.

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Department Director, Chair, Executive Management

1. May manage directly, or delegate authority to a person (or persons) in their Department the role of "Shop Manager" to manage the Shop Safety Program requirements per the guidelines in Attachment 8.
2. Assures the Faculty Advisor of a Student Organization (SO), or the Space Manager within the Department, understands their responsibility to act as a "Shop Manager" when a SO is granted space-use for conducting student-lead shop work activities within a Department.
3. Ensures any SO using Department space has a documented MOU outlining the requirements for safe use of the space using [Attachment 3](#).
4. Assures Department participation in the "Shop Safety Committee" (SSC).
5. May inform, or delegate the Shop Manager to inform, EH&S of the name, title and contact information for the person(s) representing their Department's shop(s) on the SSC using [Attachment 3](#).

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Principal Investigator / Instructor / "Supervisor" / Project Manager / Space Manager / Faculty Advisor of Student Organization(s)

(other than a "Shop Manager")

The Principal Investigator / Instructor / "Supervisor" / Project Manager / Space Manager / Faculty Advisor of a Student Organization (SO) must, or may delegate in writing, a qualified person or persons to:

1. Determine who under their direction, by name or position title, must be an "Authorized Person" to work in shop(s) by completing the "Authorized Person" qualification form (top half of [Attachment 1](#)) and provide an individual copy of this form, completed for each "Authorized Person" under their oversight, to the Shop Manager(s) where the person will work.



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2. Assure that “Authorized Person(s)” under their direction receive documented training to qualify them as an “Authorized Person” prior to allowing them access to shop equipment and work-processes.
3. Keep records of Authorized Person(s) as required in the “Record Keeping” portion of this program.
4. If operating a “ScLab”, assign who under their direction, by name or position title, must be a “ScLab Safety Coordinator”.
5. Send a completed / signed copy of MOU [Attachment 3](#) for every ScLab Safety Coordinator and SO under their purview to EH&S Shop Safety Program Manager.
6. Assure that the scope and schedule of projects under their direction allow shop users time and resources to reasonably conform to shop safety requirements as outlined in training.

Additional Duties -

1. **When hiring Contractors to conduct work in Stanford shops:** The PI / Instructor / Supervisor / Space Manager / Project Manager must assure every Contractor employee has been trained on this Shop Safety Program, been oriented to the shop / makerspace and its safety equipment / safe-work processes, and reviews / documents each Contractor employee’s level of safe-work proficiency, and any work restrictions, prior to allowing any Contractor to work in the shop’s “Restricted Area”. See [Attachment 8](#) for further information.
2. **When acting as a Space Manager and/or Faculty Advisor of a Student Organization (SO):** The act of granting access to a space for the SO to use and set-up “Shop Work Processes” indicates you are taking on the role of “Shop Manager” for the SO. This role may be delegated in writing to another person, including specific members of the SO during the ongoing use of the space, but you are responsible to ensure the requirements of this Shop Safety Program are being integrated into SO operations and followed by the SO members using the space under your oversight. This relationship is documented as a Memo of Understanding (MOU) using [MOU Attachment 3](#) and must be renewed annually.

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Student Organization (SO) Using / Managing Shops

A Student Organization that uses or builds its own shop / makerspace must be designated by the Faculty Advisor completing a Student Organization [MOU Attachment 3](#) outlining the details of the Student Organization shop. A Faculty Advisor may approve use of a Shop Space to an SO for up to one year. At the end of the year, a review and updating of the MOU must be completed for the next year. Completed MOUs should be copied to the Shop Safety Committee and EH&S Shop Safety Program Manager.

If the Faculty Advisor is designating the SO to use an existing academic shop and equipment, then the "Authorized Person" process is used for each individual member of the SO and signed-off by the Shop Manager using [Attachment 1](#).

Student Organizations:

1. Must have specific members of the SO delegated the authority to be the SO "Shop Manager" by the Faculty Advisor and/or Space Manager granting access to the space the SO will use to set up their shop area. This relationship is documented using [Attachment 3](#).
2. SO Shop Manager is responsible for all activities outlined in the "Roles / Responsibilities" of a Shop Manager as applied to the SO Shop, including all safety-management practices outlined in the "Shop / Makerspace Management Guide". (See [Attachment 8](#)).
3. SO members must follow the shop safety rules and any SOPs as set by the delegated SO Shop Manager for the space used by the SO.
4. SO members must provide the Faculty Advisor and/or Space Manager evidence of adoption and integration of this Shop Safety Program into SO operations upon request.
5. SO members provide all their own shop equipment, unless negotiated and documented in [Attachment 3](#) with supporting documentation for the use of University-owned facilities and equipment.
6. SO members operating in Stanford University owned space and/or equipment follow the same authorization / orientation process used for Stanford University personnel by completing [Attachment 1](#) and [Attachment 2](#) for each individual SO member using the space.

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ScLab Safety Coordinator

1. Works as communications liaison between a lab-space with ancillary lower-hazard shop processes (see Definition of "ShLab") and the Shop Safety Committee.
2. Ensures users of the space are aware of the hazards of shop processes in the space.
3. Conducts the annual shop self-assessment ([Attachment 7](#)) of the space to ensure safety.
4. May initiate changes and improvements to the safe work practices in the space based upon the results of the shop self-assessment.
5. Distributes safety information to space users from the Shop Safety Committee.
6. Maintains Training Records of all persons who do shop processes in the space.

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Shop Safety Committee (SSC)

1. Oversees application and operation of the Shop Safety Program
2. Works with EH&S Shop Safety Program Manager to update and direct the content of the written Shop Safety Program.
3. Shares safety information among shops across campus via an EH&S-sponsored website, a shop list-serve, EH&S / SSC Shop Safety Posters, Lessons Learned handouts, discussion at SSC meetings, and other communications means.
4. Meets roughly quarterly (e.g. four (4) times a year).
5. Identifies and manages shop safety training needs.
6. Audits Shop Safety Program through visits and learning sessions to member shops.
7. May investigate shop incidents, author and distribute "Lessons Learned" information to shops.
8. Alerts EH&S when the Shop Safety Program needs EH&S consultation / support.

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EH&S Shop Safety Program Manager

1. Is an assigned safety specialist and/or safety engineer and acts as point-person from EH&S for the Shop Safety Committee and manages all aspects of the academic Shop Safety Program for Stanford University.
2. Edits and maintains this program to meet or exceed Cal-OSHA, IIPP and other compliance requirements,
3. Informs Departments with shops of this program's requirements,
4. Provides general Shop Safety Program awareness information across campus,
5. Updates and maintains the EH&S / SSC Shop Safety website and online safety reference materials, all template-forms and attachment documents as may be needed, for implementation of the Shop Safety Program across campus, and as SSC needs may change.
6. Organizes SSC meetings and provides administrative, logistical and financial support to the SSC for ongoing Shop Safety Program management.
7. Maintains the campus' Shop Inventory and provides data to the SSC for managing shop safety across campus.
8. Maintains the SSC member-contact-list.
9. Provides assistance and training for the Shop Safety Committee, Departments, Shop Managers and "Authorized Person(s)" in implementation of this program,
10. May investigate shop incidents, author and distribute "Lessons Learned" information to shops via the Shop Safety Committee website and e-distribution list.
11. Hosts periodic meetings of the Shop Safety Committee
12. Recommends new Shop Safety equipment and processes as they are learned / developed,
13. Provides assistance in development of Shop Safety procedures, selection of PPE, etc., and
14. Reviews and updates this program periodically or as regulatory changes may dictate.

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SHOP SAFETY HAZARDS AND CONTROLS

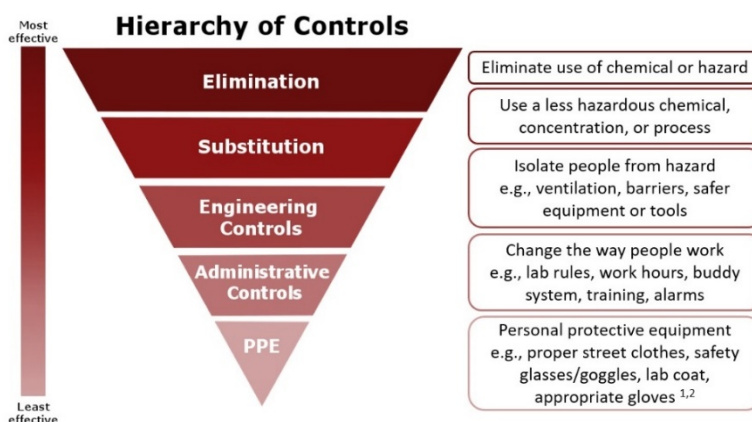
Just as in the experimental process in a lab, every shop must identify/explore hazards in the workplace, evaluate and plan for controlling those hazards, implement / challenge those hazard(s) through controls, and assess the controls to assure they are properly working. This process is ongoing and continuous as part of risk control by shop management and users. All hazard controls are applied by Shops according to the following Hierarchy of Controls.....



Hierarchy of Hazard Controls Used at Stanford University:

Eliminate the Hazard – The Most Effective Control

The most effective means to improve safety is to eliminate hazards so they cannot cause injury / loss. Shops are encouraged to eliminate hazards if practicable. (An example: Replacing a caustic chemical cleaner such as “ammonia” with bio-degradable cleaner like “Simple Green” eliminates indoor air quality and skin exposure hazards.)



Engineering Controls – Next Most Effective Control

Engineering controls are used to reduce but not necessarily eliminate a hazard. Such controls may isolate the hazard from a person being exposed to it (e.g. – machine guard) or remove the hazard when exposure may happen (e.g. – fail-safe controls with interlocks on access door panels on a punch-press).

Administrative Controls

Administrative controls are used when engineered controls / hazard elimination are not practical. Such controls rely on training, behavior and safety-program administration to assure hazard control. (Examples: Developing and following a written Standard Operating Procedure, or implementation of this Shop Safety Program into all shop work activities.)

Personal Protective Equipment (PPE) – Least Effective Control

PPE is the least effective hazard control and is only used to supplement other hazard control measures. PPE endeavors to isolate the worker from the hazard, but does nothing to reduce or eliminate the hazard. Examples of PPE include gloves, protective eyewear, hair nets, aprons, long-sleeve shirts / pants, work boots, etc... When no other means to control identified hazards are available, the Shop Manager is responsible to purchase and stock PPE appropriate to control identified hazards. PPE must be provided at no cost to all Authorized Shop Personnel and Shop Visitors entering the “Restricted Area” of a shop.

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Shop Hazards and Applying Their Controls

Eliminate Hazard (EH) / Engineer Control (EC) / Administer Control (AC) / PPE Use

Chemical Hazards	• Chemical replacement / hazard reduction (EH) • Separation of reactive chemicals (EC) • Secondary containment trays / double-walled tanks (EC) • Labeled chemical containers (AC) • Inventory of Chemicals (AC) • Hazard Communication Program (HAZCOM) (AC) • Online Safety Data Sheets (SDS) (AC) • Managing Paint and Painting Supplies (AC) • Lead, Asbestos and Silica Safety Programs (AC) • Respiratory Protection Program enrollment (AC) • JSA Review / Writing / Training (AC) • PPE Use
Compressed Gases	• Seismic Bracing (Q-Brace Program) (EC) • Cylinder Rack / Chained Storage (EC) • Energy Isolation / LOTO Program (AC) • Compressed Gas Cylinder Safety Program (AC) • Respiratory Protection Program enrollment (AC) • JSA Review / Writing / Training (AC) • PPE Use
Electrical Hazards	• Engineered Controls / Fail-safe Equipment (EC) • Special Tools (EC) • Electrical Safety Program (AC) • Energy Isolation / LOTO Program (AC) • JSA Review / Writing / Training (AC) • PPE Use
Emergency Response	• Define Egress Paths, Exits and Safe Muster Location (EC) • Hazard Communication Program (HAZCOM) (AC) • Building Emergency Evacuation Plan / Drills (AC)
Ergonomics	• Hazard reduction / elimination through design (EH) • Use / design of ergonomic tools (EC) • Fork lift / Industrial Lift / Aerial Lift Programs (AC) • Minimize repetitive work, odd postures / movements (AC) • JSA Review / Writing / Training (AC)
Heat / Cold Hazards	• Permanent insulation (EC) • Insulation Blankets (EC) • Isolation equipment / chambers (EC) • Cooling or Heating equipment (as needed) (EC) • Warning signage and direct-read temperature gages (AC) • JSA Review / Writing / Training (AC) • PPE Use
Housekeeping	• Well-planned storage and material control (EC) • "Proper" cleaning tools and vacuum equipment (EC) • Ventilation and control of sprays, dusts and fumes (EC) • Routine and institutionalized periodic Shop cleaning (AC) • JSA Review / Writing / Training (AC) • PPE Use



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Eliminate Hazard (EH) / Engineer Control (EC) / Administer Control (AC) / PPE Use

Hot Work / Fire Safety	• Fire Sprinkler System (EC) • Fire Extinguishers (EC) • Ventilation (EC) • Contain flammables in fire-safe storage cabinets (EC) • Point-of-use / small-quantity dispensing of flammables (EC) • Hot Work Permit / Program (AC) • Respiratory Protection Program enrollment (AC) • JSA Review / Writing / Training (AC) • PPE Use
Hydraulics / Fluid under pressure	• Hazard reduction through design (EC) • Use of Pressure Relief Valves (EC) • Energy Isolation / LOTO Program (AC) • Hydraulic Safety Program (AC) • JSA Review / Writing / Training (AC) • PPE Use
Material Handling	• Space layout for minimal material movement (EC) • Minimize lifting / movement work by design (EC) • Crane Safety Program (AC) • Fork lift / Industrial Lift Program / Aerial Lift Program (AC) • JSA Review / Writing / Training (AC) • PPE Use
Mechanical Equipment Hazards	• Hazard reduction through design (EC) • Machine Guards and Control Interlocks (EC) • Energy Isolation / LOTO Program (AC) • Blocks and Extension Tools • JSA Review / Writing / Training (AC) • PPE Use
Noise Hazards	• Mufflers and Isolation equipment / devices (EC) • Noise-cancellation devices (EC) • Lead sheeting and sound absorbing materials (EC) • Hearing Conservation Program (AC) • JSA Review / Writing / Training (AC) • PPE Use
Vibration	• Counter-weights and balance shafts (EC) • Cushioned / shock-proof equipment (EC) • Ergonomics Program (AC) • JSA Review / Writing / Training (AC) • PPE Use
Unfamiliar Processes / New Operations	• Research Proposal and Compliance Review Form (AC) • Hazard Identification, Evaluation and Control Program (AC) • JSA Writing / Training (AC) • PPE Use

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SHOP SAFETY PROGRAM LEADERSHIP

SHOP SAFETY COMMITTEE (SSC)

The University Committee for Health and Safety (UCHS), the President's Office representative body overseeing management of safety and risk reduction, has determined an ongoing Shop Safety Committee (SSC) best serves the campus' needs to manage shop risk. The SSC is comprised of members from Departments impacted by this program and has the responsibility to oversee, review and maintain the Shop Safety Program as it applies to campus need. The SSC is supported by EH&S administratively, technically and financially.

Shop Safety Committee (SSC) Charter

The SSC is chartered to:

1. Provide a forum for Shops to learn from each other and exchange safe work ideas and practices.
2. Develop and review technical material in support of the Shop Safety Program.
3. Assist shop Owner Departments and Shop Managers in the application of Shop Safety requirements.
4. Assess the performance of the Shop Safety Program through various means including audits, inspections, and reviews of shop accidents and near misses with development of lessons learned.
5. Disseminate "Lessons Learned" to all shops at Stanford University.
6. Meet quarterly or more often to discuss Shop Safe-work Practices and guide ongoing management of safety in academic shops.
7. Provide the EH&S Shop Safety Program Manager with recommendations for improved training and program implementation throughout campus.
8. Provide EH&S Shop Safety Program Manager with recommendations for funding of Shop Safety Committee initiatives.

SSC Composition and Function / Logistics

Permanent members of the SSC include the EH&S Shop Safety Program Manager, an Ergonomist (EH&S), the SSC Chair from an affected Department, and a volunteer-member representative from each shop on campus who are knowledgeable in shop work practices and safe-work needs of their respective shops.

The SSC is chaired by a voluntary member of the Shop Safety Committee (not EH&S). The SSC Chair lasts for 8 meetings (two years). The Chair volunteers to be the Chair and is ratified by a simple-majority vote of SSC members in attendance at the Fall-meeting of the SSC every other year. The Chair, with support of the EH&S Shop Safety Program Manager, plans the committee meeting agenda using [Attachment 5](#) and leads the meetings.

The EH&S Shop Safety Program Manager arranges for a meeting location, provides refreshments, documents committee meetings by keeping minutes, keeps attendance rosters using [Attachment 6A](#), and may edit / amend the Shop Safety Program as directed by an SSC super-majority vote (66% or greater) during a meeting, or if dictated by new compliance codes.

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SHOP SAFETY PROGRAM REQUIREMENTS

CREATION OF “RESTRICTED AREAS” and “AUTHORIZING” PERSONNEL

Only persons who are “Authorized” may work in “Restricted Areas” of shops. Any person who is not “Authorized” may only enter “Restricted Areas” of shops with continuous escort by an “Authorized Person” from that shop.

Restricted Areas

The “Restricted Area” is the physical space where any of the high-hazard shop-processes outlined in the Scope of this program are routinely conducted. This area is demarcated as separate from general-use / public areas of the shop, and is designed for high-hazard shop processes to safely and routinely be performed by Authorized Personnel.

Under the more-detailed guidance of [Attachment 8](#), the Shop Manager must determine the physical space within the shop that is the “Restricted Area”, and take measures to clearly demarcate (e.g. by signage, paint-striping, installation of barriers, locked door, etc.) these areas as ‘separate’ from other “non-restricted” shop space.

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Authorizing Personnel

A person must be “authorized” before they may conduct work in a shop. This process is conducted for each “Authorized User” by the Shop User’s Supervisor / Instructor combined with the Shop Manager (if a different person), and documented by the Shop Manager using [Attachment 1](#). An “Authorized Person” is an individual formally recognized as:

1. Having requirements for working in the Shop documented by their PI / Instructor / Supervisor in the top-half of [Attachment 1](#), and
2. Having completed required classroom, trades or other training, and
3. Having sufficient understanding of Shop safe-work practices, equipment, and facilities to be able to recognize and positively control any hazards that may be present, and
4. Possessing the work experience and formal training necessary to execute work according to recognized and accepted Shop safe-work practices, and
5. Having completed orientation to a shop’s “Restricted Area(s)”, facility, equipment and apparatus as documented by the Shop Manager on the bottom-half of [Attachment 1](#), and
6. Reviewing and signing of [Attachment 2](#) and the University’s HAZCOM Program as it may apply to a specific shop.

A person may be considered “Authorized” with respect to certain equipment and safe-work methods in some work activities or shop locations, but not ‘Authorized’ for another activity/location within the same shop. It is the responsibility of the Shop Manager to determine the degree of “Authorization” for each and every person working in that shop, and document this on the “Authorized Person’s” record by completing the bottom half of [Attachment 1](#).

Authorization to work in one shop does not allow a person to work in another shop. A person must be “authorized” in each and every shop to conduct work in that specific shop under the specific authorizing Shop Manager. Shop Managers who oversee multiple shops may authorize a person to work in more than one shop under their direction, but this must be clearly stated on the authorization documented on [Attachment 1](#) for that individual.



Safe Work Processes

SHOP SAFE WORK REQUIREMENTS

It is the requirement of Stanford University and any department that owns / manages a “Shop” / “Makerspace” that:

1. Only persons who are “Authorized” by a Shop Manager / Supervisor may work in restricted areas of shops unescorted.
2. A person becomes “Authorized” as indicated by their need to work in a shop, and approval from their immediate Supervisor, PI or Instructor combined with a Shop Manager’s agreement that they possess the training and qualifications for safe work in the shop. This is documented using Attachment 1 and Attachment 2 of the Shop Safety Program.
3. Authorization is done for each and every shop, regardless of owner. A Supervisor / PI / Instructor or “Other Shop Manager” cannot authorize access to any shop in which they are not the “Shop Manager” as designated by the Shop Owner using Attachment 3.
4. Shop Managers define and maintain a “Restricted Area” within their shop in which high-hazard shop processes occur, and that only “Authorized Persons” may enter and work in unescorted. This is also documented in Attachment 3.
5. Shop Managers determine the appropriate safe-work training needs for their shop, and assure that all “Authorized Persons” in the shop have received the orientation and training to their shop. At minimum this includes reading and signing Attachment 2 of Stanford’s Shop Safety Program.
6. Stanford University Shops support a Stop Work Rule when unsafe work is witnessed. It is the responsibility of every Shop User to exercise this rule when unsafe work conditions or practices are observed. If you see unsafe behavior or activities, don’t hesitate to use the Stop Work Rule, immediately speak up and stop unsafe work.
7. Stanford University Shops comply with Cal/OSHA regulations, the California Electrical Code and other established safety standards by applying this and other EH&S Safe-work Programs / Practices / Requirements as listed in the “Typical Shop Hazards and their Controls” section of the “Stanford University Shop Safety Program” on the EH&S website.
8. All electrically energized shop equipment is used in a safe manner as intended by the manufacturer and within the equipment’s Nationally Recognized Testing Laboratory (NRTL) functional listing.
9. Shop “Authorized Users” must resist the temptation to rush, and supervisors / instructors / PIs / shop manager must refrain from applying pressure on users to rush work.

SAFE WORK PROCEDURES

USE OF PERSONAL PROTECTIVE EQUIPMENT (PPE)

All Shop Managers are required to conduct hazard assessments of operations in their shops and apply the “Hierarchy of Hazard Controls” to controlling hazards in their shops. When the use of PPE is required, PPE must be stocked and supplied by the Shop Manager at no cost to all



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“Authorized Users” and “Visitors” who enter a “Restricted Area”. Selection is based upon evaluation of hazards that are present as follows:

1. Appropriate face and/or eye protection whenever powered equipment or flying / airborne particles or hazardous substances or injurious light sources are present.
2. Hair-nets or caps when long hair could get wrapped into rotating or rolling equipment, or could be exposed to flames or other such hazards.
3. Hard hats or bump caps when someone is working over your head, head-banger obstructions exist, or when a Shop Manager / Supervisor requires them.
4. Dust masks may be used by anyone at any time for airborne dust, but use of other respirators is determined by indoor-air-quality hazard assessment and are distributed to individual personnel through their enrollment in the Stanford University EH&S Respiratory Protection Program.
5. Gloves and other hand protection must be selected and worn based upon hazards. Use EH&S's Glove Selection Guideline for ideas on what kinds of gloves should be worn based upon hazards.
6. Full-cover shoes with non-slip soles and an enclosed heel are required in all “Restricted Areas”. Safety boots, optionally with steel / composite toes, are required during the movement of heavy equipment and work materials, or the use of industrial lift equipment or cranes.

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PERSONAL SAFETY

1. Long and medium length hair, including facial hair, must be tied back and put into a bun or inserted under a shirt-collar, or contained in a hair-net, when entering and working in a “Restricted Area” of any shop.
2. Open-toed shoes, flip-flops, clogs, sandals, shorts, scarves and loose-fitting clothing ARE NEVER WORN in any shop’s “Restricted Area”.
3. No food or drink is to be brought into and/or consumed in a “Restricted Area” of any shop.
4. Remove all dangling or bulky jewelry before working in a “Restricted Area”. Best practice is to remove all jewelry.

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EQUIPMENT OPERATION

The operation of any and all shop equipment requires prior training and approval by the Shop Manager. Only persons authorized by the Shop Manager and trained in the safe operation of the shop equipment are permitted to work in a shop and operate the specific shop equipment they have been trained on. A signed “Authorization Form” ([Attachment 1](#)) certifies this level of qualification and must be on file with the Shop Manager for each shop user. Failure to adhere to this requirement will result in a loss of shop-use privileges.

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SOLO WORK PRACTICES

The **PRIMARY HAZARD OF SOLO WORK** is sudden injury or illness without anyone nearby who can assist with the prompt initiation of emergency response procedures. In addition, after-hours work lacks the most-rapid access to the most-trained campus emergency responders. Considering



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an emergency situation when every second counts, anyone contemplating working alone in a shop must address the following General Solo Work Rules prior to doing so.

GENERAL SOLO WORK RULES IN ACADEMIC / STUDENT SHOPS

1. Shop access outside of supervised hours is restricted to specific "Authorized Persons" based upon the Shop Manager's discretion. Access for on-going solo work is not granted in any Academic Shop. On-going and/or routine working solo outside of 'normally scheduled operation and/or staffed hours' is strictly forbidden and may result in revocation of shop-use privileges by the shop's owner department.
2. An "Authorized Person" may work solo in a Shop during 'normally scheduled operation and/or staffed hours' when communications means are made with the Shop Manager / Supervisor to assure the person's continuous safety and ability to communicate immediately with emergency responders.
3. When using large fixed powered shop equipment, flammable solvents, acids and/or etchants, heat and/or spark-producing equipment, at least one other authorized person is to be in the shop area.
4. Provisions for solo-work on a case-by-case basis can be made between the shop manager and the solo worker by completing a "Solo Work Risk Mitigation Plan" (Attachment 4) as outlined in that section of the Stanford Shop Safety Program document.

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RISK MITIGATION PLAN FOR SPECIFIC SOLO WORK

When solo work must be conducted outside of "normally schedule operations and/or staffed hours", a Risk Mitigation Plan ([Attachment 4](#)) must be completed and documented by the Solo Worker, in consultation with and approval by the Shop Manager, that address the following:

1. Requests to work alone after hours, or on weekends, must be approved by the Shop Manager case-by-case each time solo work is to be performed.
2. Consult with the Shop Manager when planning to work alone. What operations will be conducted, and what means for immediate communication in an emergency are needed during solo work?
3. Development of the Solo Work Risk Mitigation Plan must include review of the specific work activities, associated hazards and hazard controls to be used; and an emergency communication plan including contact name(s) and contact method(s).
4. If possible, solo workers should communicate with others in the building when working alone in the shop. Let them know when you arrive, what you're doing and when you leave.
5. Always avoid performing any high-hazard operations when working alone in the shop.
6. Never leave shop equipment unattended and operating.

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SHOP MANAGER'S CONSIDERATIONS FOR MANAGING SAFER SOLO WORK

Shop Managers must consider requests to work solo on a case-by-case basis. Any combination of the following must be considered to ensure control of the hazards of solo work to an effective degree:

1. The authorized shop user's training and experience.



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2. Tasks and hazards involved in the planned solo work.
3. The likelihood of an injury or illness that would prevent the shop user from calling for help.
4. Consequences resulting from a worst-case scenario of sudden injury or illness.
5. Deadlines, time-constraints and harried pressure to complete work that could negatively affect the shop user's safe work-methods and clear-headed work-approach.
6. The shop user's physical condition and any constraints that may impact their safe work approach and methods. [Consult with local Human Resources Officer for guidance and compliance with the Americans with Disabilities Act (ADA)].
7. The time of day solo work is to be conducted, for example, during normal business hours (7 am – 6 pm Monday through Friday) versus at night or weekends/holidays.
8. The length of time the solo work is to be done (e.g. 5 minutes vs 5 hours vs 5 days)
9. What Personal Protective Equipment should be worn during solo work?
10. What Emergency Response Procedures, equipment and/or contact information is needed for solo work to take place? Besides the solo worker, who needs access to this information and equipment? How will they get this information prior to work commencing?
11. Safe and unobstructed access to the nearest eyewash, shower, and fire extinguisher.

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RISK MITIGATION PLAN OPTIONS

The shop manager in conjunction with the solo worker may choose to implement any combination of the following risk-reduction options to control the hazards of solo work. These strategy options are to be implemented by honor system and mutually agreed to between Shop Manager and the Authorized Person on a case-by-case basis prior to the shop user being allowed to work solo in an academic shop:

HAZARD REDUCTIONS

1. Always wear proper PPE
2. Only hand-tool use is allowed when working solo.
3. Limit higher-hazard solo work to normal week-day business hours when other personnel are likely to be nearby and able to respond to an emergency.
4. Substitute low-hazard chemicals for high hazard chemicals while doing solo work. (e.g., use water-based paint instead of oil-based paint and solvents)
5. Avoid setting up large-scale/hazardous projects when working solo.
6. Adopt Shop Rules so that No High-Hazard Work is done solo nights and weekends, including:
7. Hazardous Chemical Use (Solvents, Etchants, Acids, Fuels, etc...)
8. Powered Machinery use
9. Hand-held Power Tools use
10. Heat / Spark / Arc / Open Flame / Smoke producing operations

ADMINISTRATIVE CONTROLS

1. Set up a Buddy system – In person OR by cell phone / radio. Either way, your buddy must be able to recognize trouble and seek emergency help without the shop user's assistance.
2. Develop and share specific emergency procedures with your buddy using this "Solo Work Risk Reduction Plan".
3. Install an "emergency response" / "person down" app on your mobile device that will automate the call-out function in the event the solo worker stops moving or becomes horizontal for an extended period of time.



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4. Communicate with others in the building, building security, campus police or the shop manager. Let them know when you arrive, what you're doing and how long you intend to be there, and when you leave.
5. Have your buddy check in on you every 10-15 minutes, OR....Check-in with your buddy every 10-15 minutes.
6. Keep shop doors open (when there's no door window) or unlocked (with a door window) when solo work is happening.
7. Setup your buddy in a safe place in the shop or shop-adjacent so they can be present and aware of your wellbeing if you must do high-hazard solo shop work.

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SHOP SAFE WORK PRACTICES

COMPRESSED AIR: Compressed air or other gases in excess of 10 pounds per square inch are never used to blow dirt, chips or dust from clothing while it is being worn, or from any surface as part of routine shop cleaning. Use suction devices or other cleaning methods instead.

DRAIN DISPOSAL: Hazardous waste may not be poured down the shop drain. The only materials that may be disposed down the drain are in the [Non-hazardous chemical wastes list](#)*. All hazardous and chemical wastes must be disposed of by EH&S. Failure to comply with local permit requirements for campus drain disposal could lead to substantial fines. For more information see the Stanford Water Planning and Stewardship [SOP](#)** and the document on [Best Management Practices](#)***.

*< <https://ehs.stanford.edu/reference/list-non-hazardous-chemical-wastes>>

**< https://suwater.stanford.edu/sites/g/files/sbiybj7756/f/glass_wash_sop.pdf>

***< https://suwater.stanford.edu/sites/g/files/sbiybj7756/f/wastewater_bmp_factsheet.pdf>

ELECTRICAL SAFETY: All work on and development of electrical equipment must be conducted according to guidelines and compliance requirements set by NFPA70 National Electrical Code, or the California Electrical Code if more stringent. Work that must be done on energized electrical circuits must be done in accordance with NFPA70E requirements. Otherwise, work on electrical equipment must be done verified de-energized and locked / tagged out by all persons working on the equipment.

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EMERGENCY PREPAREDNESS AND RESPONSE:

- Always consider electrical equipment, pressurized systems, mechanical components or compressed gas equipment "energized" until you yourself have positively proven it otherwise.
- Do not take someone else's word that they turned off the electric power, bled pressurized systems to zero pressure, or blocked mechanical systems from moving. Always verify for yourself that equipment is properly de-energized and movement is stopped before removing machine guards or working in high-hazard areas of shop equipment.
- Learn and know how to shut down equipment in an emergency BEFORE there's an emergency.
- Design and plan for safety by looking for hazards and anticipating problems.
- Always de-energize the equipment before testing or working on it.



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- Positively ensure all hazards are identified before blocking, blinding, bleeding or conducting lock-out and tag-out.
- Lock-out and Tag-out broken, mothballed or unsafe equipment until it can be repaired or permanently removed.
- Refer to the shop's Hazard Communication (HAZCOM) Program and Safety Data Sheets (SDS) to understand material / chemical hazards, emergency contact information, and emergency response procedures.

ENERGY ISOLATION – LOCK OUT TAG OUT: Know and apply the Energy Isolation / Lock Out - Tag Out process to all facility and equipment locations whenever:

- Hazardous energy release could cause unexpected movement of equipment,
- Unexpected flow of energy could cause injury or equipment damage,
- A machine-guard or interlock is removed or bypassed, or
- There is more than one energy supply to a machine, and the energy source cannot be simply 'unplugged' so that energy-control is exclusive to the person working on/with the equipment.

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FORKLIFTS / AERIAL LIFTS: Only drivers / operators authorized by the Shop Manager and enrolled in the Stanford Industrial Lift Truck Safety Program and/or Aerial Lift Safety Program who hold a current operator's license issued by the EH&S Program Manager are permitted to operate such vehicles.

HAND TOOLS AND SHOP EQUIPMENT: Only persons authorized by the Shop Manager and trained in the safe use of hand tools and shop equipment are allowed to use them. Use hand tools and shop equipment only as they were designed / intended to be used.

HOUSEKEEPING: The work and surrounding area in which you are working must be kept neat, clean and safe at all times. If you see a mess, spilled liquid or debris, whether you created it or not, take the time to clean it up.

HYDRAULICS: Only persons authorized by the Shop Manager and trained in the safe operation and/or development of hydraulic systems are permitted to assemble, qualify, and operate hydraulic equipment connected to any shop hydraulic system.

OVERHEAD CRANES / HOISTS: Only persons authorized by the Shop Manager and trained on the Campus Crane Safety Program and who are deemed a "Qualified Crane Operator" according to the requirements of the program are permitted to operate such equipment.

WORK BRIEFING: Participate in a safe work briefing with all personnel in the area before commencing any shop work. Assure you discuss planned work with the Shop Manager, or their designee, to identify any hazards of your work, or other hazardous conditions nearby due to work being done by others. Know the Shop's emergency procedures. Maintain the protection of covers, barriers and shielding of all electrical equipment, hot work, machine guards, chemical storage, etc. Never drill into a wall or floor slab without Shop Manager **and** Building Manager approval.

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SHOP SAFETY PROGRAM TRAINING REQUIREMENTS

Faculty, Staff and Students

All Faculty, Staff, Students and Contractors must know that under no circumstances is anyone allowed to conduct work in a “Shop” without prior “Authorization” as defined in this program document. When anyone receives training associated with this Shop Safety Program, they must sign a training roster ([Attachments 6A and 6B](#)) issued by the instructor / trainer / Shop Manager.

“Authorized Person”

PI / Instructor / Supervisor, and the Shop Manager must jointly assure the knowledge of any “Authorized Person” and provide them training as needed to assure safe work in the shop. The “Authorized Person” must be familiar with the contents of this program, be able to recognize hazards and know what control measures are to be implemented to control shop hazards. At minimum, they must know the Shop Safe Work Requirements, Practices and Procedures as outlined in [Attachment 2](#).

Principal Investigator / Instructor / “Supervisor”

(other than a “Shop Manager”)

The Principal Investigator / Instructor / “Supervisor” must know their roles and responsibilities in the application of this program to their work and people reporting to them, and assure compliance with this program by all who fall under their responsibility.

Shop / Makerspace Manager

The Shop Manager must know their roles and responsibilities in the application of this program to their work, be trained on Shop Management as may be appropriate to ensure safety in location(s) and operation(s) under their purview, assure proper documented training of people reporting to them and other “Authorized Persons” working in the shop, and assure compliance with this program by all who fall under their responsibility.

Department Director / Chair / Executive Management

The Department Director / Chair must know their roles and responsibilities in the application of this program to their work, and support the implementation of this program’s requirements into the business / operations of their Department.

ShLab Safety Coordinator

The ShLab Safety Coordinator is trained on the contents of this program and how it may apply to work and equipment used in their lab. They are also trained on Shop Safety Program “Record-keeping” responsibilities, management of the annual shop self-assessment program in their lab, and other duties as outlined in the “Roles and Responsibilities” section.

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Contractors and Joint Projects in Shops

Contractors follow their own Shop Safety Program and procedures and must train their personnel when using their own equipment and shop processes. The Contractor provides the University evidence of their Shop Safety Program upon request. The Contractor provides all their own shop equipment unless negotiated by contract with the University for use of University equipment.

For Joint Projects where employees of the University and Contractor(s) are working in the same shop at the same time, the Shop Manager is in control of the shop and must hold joint "tailgate" meetings with all personnel in attendance who will be working in the area to promote understanding of safe work practices, and open lines of communication amongst workers. Contractors using University equipment and property follow University safe-work procedures and review and follow JSAs and SOPs as developed by the University as part of their training. Contractor training conducted by Stanford University is documented by the Shop Manager using a training roster ([Attachment 6B](#)).

Shop Safety Committee member

SSC members are trained on the Committee Charter, the Committee Roles and Responsibilities, the Shop Safe Work Requirements, Practices and Procedures, and other Shop Safety Program requirements. They receive this training by review of this program document and discussion with other Committee members.

EH&S

Provides training and guidance to the Shop Safety Committee as needed to keep program momentum moving forward, and the program current to compliance requirements and campus need.

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RECORD KEEPING REQUIREMENTS

PIs / Instructors / Supervisors keep the following documentation:

- Completes top half of [Attachment 1](#) for each person they determine must become an "Authorized Person" to work in a shop. Copy of the completed top-half must be kept for duration of employment, tenure or work-study term plus 3 years.
- Copies of any training rosters ([Attachment 6B](#)) and training content. Keep for 3 years.
- Copies of any procedures / protocols developed for shop safety. Keep for as long as they are current.

Shop Manager keeps the following documentation:

- Completes bottom half of [Attachment 1](#) for each person who must become an "Authorized Person" to work in a shop. If the PI / Instructor / Supervisor is also the Shop Manager, completes both top and bottom halves of Attachment 1 for each authorized shop user. This must be kept for duration of employment, tenure or work-study term plus 3 years.
- For Student Shops, use Attachment 1 for a duration of Authorization for shop use for a semester up to a maximum of 1 year. At the end of 1 year, the Student may be re-authorized for ongoing shop use of up to another year if their academic program dictates need. This must be kept for duration of academic program plus 3 years.
- A signed copy of page 3 of the "Shop Safety Policy" ([Attachment 2](#)) given to each and every "Authorized Person" working in their shop. This must be kept for duration of employment, tenure or work-study term, or academic program duration, plus 3 years.
- Copies of any training rosters ([Attachment 6B](#)) and training content. Keep for 3 years.
- Copies of any procedures / protocols developed for shop safety. Keep for as long as they are current.

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ShLab Safety Coordinators keep the following documentation:

- Authorized Person(s) who conduct shop-processes in labs documents ([Attachment 1](#) and [Attachment 2](#)) for the duration of their lab work plus 3 years.
- A current [MOU Attachment 3](#) for the Lab.
- Any training rosters ([Attachment 6B](#)) for 3 years.
- Any Safe-work Protocols / SOPs / Other Procedures as long as they are current.
- Any JSAs that are developed by the Department to address Shop Safety concerns for as long as they are current.
- "Shop Self Assessment" forms ([Attachment 7](#)) completed for the lab for the previous two years.

EH&S keeps the following documentation:

- Master copy of the Shop Safety Program and maintains this program to meet or exceed Cal-OSHA requirements.
- SSC meeting minutes and posts them on the SSC website (maintained by EH&S)
- General program awareness and information dissemination across campus.
- Training content and rosters for any training completed by EH&S with copies provided to Shop Safety Committee, Departments, Labs and Shop Managers as may be necessary.



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- Provides and maintains the Hazard Communication (HAZCOM) program templates that are customized and posted by Shop Managers for each shop.
- Any documents relating to shop incident investigations and “Lessons Learned” information to Shops via the Shop Safety Committee website and e-distribution list.
- Any consultations / recommends for Shop Safety, and
- Program updates as need may dictate.

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REFERENCES AND RESOURCES

California Title 8 – Cal-OSHA General Industry Code of Regulations

- **Compressed Gas and Air Equipment** <http://www.dir.ca.gov/Title8/sb7g9.html>
<http://www.dir.ca.gov/title8/3301.html>
<http://www.dir.ca.gov/title8/4650.html>
- **Cranes and Hoisting Equipment** <http://www.dir.ca.gov/Title8/sb7g13.html>
- **Ergonomics** <http://www.dir.ca.gov/Title8/5110.html>
- **Fire Safety** <http://www.dir.ca.gov/Title8/sb7g27.html>
- **Flammable Liquids, Gases and Vapors** <http://www.dir.ca.gov/Title8/sb7g20.html>
- **Hand and Portable Power Tools** <http://www.dir.ca.gov/Title8/sb7g3a20.html>
- **Hazardous Energy – Lock out Tag out** <http://www.dir.ca.gov/title8/3314.html>
<http://www.dir.ca.gov/dosh/etools/08-003/LockoutSteps.pdf>
<http://www.dir.ca.gov/dosh/etools/08-003/P08-00302.pdf>
- **Hazardous Materials** <http://www.dir.ca.gov/Title8/sb7g16.html>
- **Housekeeping / Work Area Conditions** <http://www.dir.ca.gov/Title8/sb7g1a4.html>
<http://www.dir.ca.gov/Title8/sb7g1.html>
- **Machine Guarding** <http://www.dir.ca.gov/Title8/sb7g8.html>
<http://www.dir.ca.gov/Title8/sb7g6a37.html>
<http://www.dir.ca.gov/Title8/sb7g6.html>
- **Noise** <http://www.dir.ca.gov/Title8/sb7g15a105.html>
- **Personal Protection (PPE) / Safeguards** <http://www.dir.ca.gov/Title8/sb7g2a10.html>
- **Safe Work Practices** <http://www.dir.ca.gov/Title8/sb7g2a7.html>
- **Vehicles and Shop Mobile Equipment** <http://www.dir.ca.gov/Title8/sb7g4.html>
- **Welding / Cutting Operations – Electric** <http://www.dir.ca.gov/Title8/sb7g11.html>
- **Welding / Cutting Operations - Gas** <http://www.dir.ca.gov/Title8/sb7g10.html>

Federal OSHA:

- Machine Controls Interlocks and Fail-safe Design Guideline document:
http://www.osha.gov/OshDoc/Directive_pdf/CPL_02-00-147.pdf

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ISSUE BY / REVIEW DATE

This program is ratified by the University Committee for Safety and Health under the direction of the President of the University.

It is issued / adopted on October 20, 2019 by Russell Furr, Associate Vice Chancellor, EH&S, Stanford University.

This program is reviewed and updated every 3 years, or as needed, by Shop Safety Committee, the EH&S Occupational Health and Safety Team, and the EH&S Shop Safety Program Manager.

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ATTACHMENTS

[Attachment 1: "Authorized Person" qualification for Shop Work – Template](#)

[Attachment 2: Shop Safe Work Requirements, Practices and Procedures - Template](#)

[Attachment 3: Academic Shop Space Designation / MOU / Identification - Form](#)

[Attachment 4: Solo Work Risk Reduction Plan](#)

[Attachment 5: Shop Safety Committee Meeting Agenda Template](#)

[Attachment 6: Shop Safety Self-assessment Template](#)

[Attachment 7: Shop Safety Training Roster – Template Form](#)

[Attachment 8: Shop / Makerspace Set-up and Management Guide](#)

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ATTACHMENT 1 – SHOP USER AUTHORIZATION

Stanford University – Academic Shop Safety Program

Attachment 1: “Authorized User” qualification for Shop Work

To: Personnel / Student File for _____ (Shop User's name)

From: _____ Date: _____
(PI / Supervisor name)

This person is (check one): ☐ STAFF / FACULTY ☐ STUDENT** ☐ VISITOR / CONTRACTOR**

** Student / Visitor / Contractor Authorization is limited to **one year or less** from date above. Duration of limited authorization must be described below by PI / Supervisor / Instructor authorizing shop access / work.

COMPLETED BY THE PI / SUPERVISOR / INSTRUCTOR OF THE “AUTHORIZED PERSON”:

Re: This document confirms required qualification of the above named person to perform:
(Check all that apply) (Attach additional pages if more space is needed)

☐ Conduct operations and work in the following Shops [List shop name(s) & Location(s)]:

☐ Operate and conduct the following shop-work processes:

☐ Date Limit or Other Restrictions of Authorization (describe):

☐ Conduct shop processes and/or operate equipment per the attached list (attach list)

If, for any reason, as their PI / Instructor / Supervisor, I think that this person is not performing shop work safely, this qualification will be revoked by me. Below are signature(s) of responsible person(s) verifying training, experience and/or providing instruction ensuring this person's knowledge of safe work skills:

PI / Supervisor Signature: _____ Date: _____

COMPLETED BY THE SHOP MANAGER / SHOP SUPERVISOR:

This designation of “Authorized Person” is based on evidence of safe performance of all duties related to Shop Safety and verification by the “Shop Manager / Supervisor” through:
(Check all that apply)

- ☐ **Experience** – This Stanford employee has received and signed Attachment 2, and demonstrated safe performance of shop work processes and skill in safe operation of Shop equipment for _____ years.
- ☐ **Previous Training** – Attachment 2 and other training¹ (includes skill checks or tests) are attached.
- ☐ **Instruction** – This shop user has received and signed Attachment 2, as well as additional instruction from me or other shop employees who are qualified. This person's work has been observed while performing shop operations. It's confirmed that this person has the knowledge to perform shop work safely.

Shop Manager/Supervisor: _____ Date: _____

If, for any reason, as the Shop Manager / Shop Supervisor, this person is not performing shop work safely, this Authorization will be revoked for use of my shop, regardless of the person's qualification above by others.

Authorized Person: _____ Date: _____

CC: PI / Supervisor file Shop User's Personnel File Shop Manager/Supervisor file

File: SHP-USER-Attachment-1-QualAuth-Shop-User-FINAL-10-18-2019.docx

¹ If training is part of the evidence used to qualify this shop user, attach records relevant to this qualification in the shop's personnel file for this person.

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ATTACHMENT 2 – SHOP USER ORIENTATION

Stanford University– Shop Safety Program

Attachment 2: SHOP SAFE-WORK REQUIREMENTS, PROCEDURES AND PRACTICES

SHOP SAFE WORK REQUIREMENTS

It is the requirement of Stanford University and any department that owns / manages a "Shop" / "Makerspace" that:

- Only persons who are "Authorized" by a Shop Manager / Supervisor may work in restricted areas of shops unescorted.
- A person becomes "Authorized" as indicated by their need to work in a shop, and approval from their immediate Supervisor, PI or Instructor combined with a Shop Manager's agreement that they possess the training and qualifications for safe work in the shop. This is documented using Attachment 1 and Attachment 2 of the Shop Safety Program.
- Authorization is done for each and every shop, regardless of owner. A Supervisor / PI / Instructor or "Other Shop Manager" cannot authorize access to any shop in which they are not the "Shop Manager" as designated by the Shop Owner using Attachment 3.
- Shop Managers define and maintain a "Restricted Area" within their shop in which high-hazard shop processes occur, and that only "Authorized Persons" may enter and work in unescorted. This is also documented in Attachment 3.
- Shop Managers determine the appropriate safe-work training needs for their shop, and assure that all "Authorized Persons" in the shop have received the orientation and training to their shop. At minimum this includes reading and signing Attachment 2 of Stanford's Shop Safety Program.
- Stanford University Shops support a Stop Work Rule when unsafe work is witnessed. It is the responsibility of every Shop User to exercise this rule when unsafe work conditions or practices are observed. If you see unsafe behavior or activities, don't hesitate to use the Stop Work Rule, immediately speak up and stop unsafe work.
- Stanford University Shops comply with Cal/OSHA regulations, the California Electrical Code and other established safety standards by applying this and other EH&S Safe-work Programs / Practices / Requirements as listed in the "Typical Shop Hazards and their Controls" section of the "Stanford University Shop Safety Program" on the EH&S website.
- All electrically energized shop equipment is used in a safe manner as intended by the manufacturer and within the equipment's Nationally Recognized Testing Laboratory (NRTL) functional listing.
- Shop "Authorized Users" must resist the temptation to rush, and supervisors / instructors / PIs / shop manager must refrain from applying pressure on users to rush work.

USE OF PERSONAL PROTECTIVE EQUIPMENT (PPE)

All Shop Managers are required to conduct hazard assessments of operations in their shops and apply the "Hierarchy of Hazard Controls" to controlling hazards in their shops. When the use of PPE is required, PPE must be stocked and supplied by the Shop Manager at no cost to all "Authorized Users" and "Visitors" who enter a "Restricted Area". Selection is based upon evaluation of hazards that are present as follows:

- Appropriate face and/or eye protection whenever powered equipment or flying / airborne particles or hazardous substances or injurious light sources are present.
- Hair-nets or caps when long hair could get wrapped into rotating or rolling equipment, or could be exposed to flames or other such hazards.
- Hard hats or bump caps when someone is working over your head, head-banger obstructions exist, or when a Shop Manager / Supervisor requires them.
- Dust masks may be used by anyone at any time for airborne dust, but use of other respirators is determined by indoor-air-quality hazard assessment and are distributed to individual personnel through their enrollment in the Stanford University EH&S Respiratory Protection Program.
- Gloves and other hand protection must be selected and worn based upon hazards. Use EH&S's Glove Selection Guideline for ideas on what kinds of gloves should be worn based upon hazards.
- Full-cover shoes with non-slip soles and an enclosed heel are required in all "Restricted Areas". Safety boots, optionally with steel / composite toes, are required during the movement of heavy equipment and work materials, or the use of industrial lift equipment or cranes.

PERSONAL SAFETY

- Long and medium length hair, including facial hair, must be tied back and put into a bun or inserted under a shirt-collar, or contained in a hair-net, when entering and working in a "Restricted Area" of any shop.
- Open-toed shoes, flip-flops, clogs, sandals, shorts, scarves and loose-fitting clothing ARE NEVER WORN in any shop's "Restricted Area".
- No food or drink is to be brought into and/or consumed in a "Restricted Area" of any shop.
- Remove all dangling or bulky jewelry before working in a "Restricted Area". Best practice is to remove all jewelry.

SOLO WORK RULES IN ACADEMIC / STUDENT SHOPS

1. Shop access outside of supervised hours is restricted to specific "Authorized Persons" based upon the Shop Manager's discretion. Access for on-going solo work is not granted in any Academic Shop. On-going and/or routine working solo outside of 'normally scheduled operation and/or staffed hours' is strictly forbidden and may result in revocation of shop-use privileges by the shop's owner department.
2. An "Authorized Person" may work solo in a Shop during 'normally scheduled operation and/or staffed hours' when communications means are made with the Shop Manager / Supervisor to assure the person's continuous safety and ability to communicate immediately with emergency responders.
3. When using large fixed powered shop equipment, flammable solvents, acids and/or etchants, heat and/or spark-producing equipment, at least one other authorized person is to be in the shop area.
4. Provisions for solo-work on a case-by-case basis can be made between the shop manager and the solo worker by completing a "Solo Work Risk Mitigation Plan" (Attachment 4) as outlined in that section of the Stanford Shop Safety Program document.



Shop Safety Program Stanford University

Stanford University– Shop Safety Program

SHOP SAFE-WORK PRACTICES

COMPRESSED AIR: Compressed air or other gases in excess of 10 pounds per square inch are never used to blow dirt, chips or dust from clothing while it is being worn, or from any surface as part of routine shop cleaning. Use suction devices or other cleaning methods instead.

DRAIN DISPOSAL: Hazardous waste may not be poured down the shop drain. The only materials that may be disposed down the drain are in the [Non-hazardous chemical wastes list](#)*. All hazardous and chemical wastes must be disposed of by EH&S. Failure to comply with local permit requirements for campus drain disposal could lead to substantial fines. For more information see the Stanford Water Planning and Stewardship [SOP](#)** and the document on [Best Management Practices](#)***.

*< <https://ehs.stanford.edu/reference/list-non-hazardous-chemical-wastes>>

**< https://suwater.stanford.edu/sites/g/files/sbiybj7756/f/glass_wash_sop.pdf>

***< https://suwater.stanford.edu/sites/g/files/sbiybj7756/f/wastewater_bmp_factsheet.pdf>

ELECTRICAL SAFETY: All work on and development of electrical equipment must be conducted according to guidelines and compliance requirements set by Stanford University's Electrical Safety Program. Work that must be done on energized electrical circuits must be done in accordance with NFPA70E requirements. Otherwise, work on electrical equipment must be done verified de-energized and locked / tagged out by all persons working on the equipment.

EMERGENCY PREPAREDNESS AND RESPONSE:

- Always consider electrical equipment, pressurized systems, mechanical components or compressed gas equipment "energized" until you yourself have positively proven it otherwise.
- Do not take someone else's word that they turned off the electric power, or bled pressurized systems to zero pressure. Always verify for yourself that equipment is properly de-energized and stopped movement before removing machine guards or working in high-hazard areas of shop equipment.
- Learn and know how to shut down equipment in an emergency BEFORE there's an emergency.
- Design and plan for safety by looking for hazards and anticipating problems.
- Whenever possible de-energize the equipment before testing or working on it.
- Positively ensure all hazards are identified before blocking, blinding, bleeding or conducting lock-out and tag-out.
- Lock-out and Tag-out broken, mothballed or unsafe equipment until it can be repaired or permanently removed.
- Refer to the shop's Hazard Communication (HAZCOM) Program and Safety Data Sheets (SDS) to understand material / chemical hazards, emergency contact information, and emergency response procedures.

ENERGY ISOLATION – LOCK OUT TAG OUT: Know and apply the Stanford University Energy Isolation / Lock Out - Tag Out Program to all facility and equipment locations whenever:

- Hazardous energy release could cause unexpected movement of equipment,
- Unexpected flow of energy could cause injury or equipment damage,
- A machine-guard or interlock is removed or bypassed, or
- There is more than one energy supply to a machine, and the energy source cannot be simply 'unplugged' so that energy-control is exclusive to the person working on/with the equipment.

FORKLIFTS / AERIAL LIFTS: Only drivers / operators authorized by the Shop Manager / Supervisor and enrolled in the Campus Industrial Lift Truck Safety Program and/or Aerial Lift Safety Program who hold a current operator's license issued by the EH&S Program Manager are permitted to operate such vehicles.

HAND TOOLS AND SHOP EQUIPMENT: Only persons authorized by the Shop Manager / Supervisor and trained in the safe use of hand tools and shop equipment are allowed to use them. Use hand tools and shop equipment only as they were designed / intended to be used.

HOUSEKEEPING: The work and surrounding area in which you are working must be kept neat, clean and safe at all times. If you see a mess, spilled liquid or debris, whether you created it or not, take the time to clean it up.

HYDRAULICS: Only persons authorized by the Shop Manager / Supervisor and trained in the safe operation and/or development of hydraulic systems are permitted to assemble, qualify, and operate hydraulic equipment connected to any shop hydraulic system.

OVERHEAD CRANES / HOISTS: Only persons authorized by the Shop Manager / Supervisor and trained on the Campus Crane Safety Program and who are deemed a "Qualified Crane Operator" according to the requirements of the program are permitted to operate such equipment.

WORK BRIEFING: Participate in a safe work briefing with all personnel in the area before commencing any shop work. Assure you discuss planned work with the Shop Manager / Supervisor, or their designee, to identify any hazards of your work, or other hazardous conditions nearby due to work being done by others. Know the Shop's emergency procedures. Maintain the protection of covers, barriers and shielding of all electrical equipment, hot work, machine guards, chemical storage, etc. Never drill into a wall or floor slab without Shop Manager and Building Manager approval.



Shop Safety Program Stanford University

Stanford University– Shop Safety Program

SHOP NAME: _____

LOCATION / DEPARTMENT: _____

I have read, had an opportunity to ask questions about, understand and been given a copy of Stanford University's "Shop Safe Work Requirements, Procedures and Practices". I pledge to only conduct work according to the safe requirements, procedures and practices outlined within the contents of the document. I understand that failure to follow requirements, procedures and practices outlined within the document may result in immediate expulsion and possibly permanent revocation of privileges to work in the shop noted above. In addition, other disciplinary action may be brought by the department and/or Stanford University management for failure to follow these requirements, procedures and practices.

Signature: X _____
(Authorized Shop User)

Print name: _____

Date: _____

Signature: X _____
(Shop Supervisor / Manager)

Print name: _____

Date: _____

Instructions to the Shop Manager / Supervisor: Please separate the 2-page "Shop Safety Requirements / Procedure / practices document and provide it as a 'hand-out' copy to every 'Authorized User' who will be working in your shop. Have each person sign / date this third page of the requirements / procedures / practices document and keep their signed page in your file as evidence of basic safety training / orientation to your shop. To save resources, it's suggested pages 1 and 2 be printed front and back on one sheet of paper.

Provide additional training as you may determine is needed based upon assessment of your shop's hazards, equipment and operations. Include a checklist of training subjects and materials covered as evidence of your training efforts.

This is Page 3 of 3 – Keep the signed copy on file as evidence of your shop safety training / orientation of each Authorized Shop User.

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Shop Safety Program Stanford University

ATTACHMENT 3 – SHOP SPACE / MOU / SET-UP INFORMATION

Stanford University Shop Safety Program – Attachment 3 Academic Shop / Makerspace - Space Designation / MOU / Safety Contact Identification

Purpose: This form documents the designation of physical space for Academic Shop and Maker-Space work processes. It identifies key personnel and information about the space to ensure safety of occupants. It may be applied when the activities conducted in the space fall under the scope of the Shop Safety Program. It may also be used as an MOU between a Faculty Member (as their approval for space use) and a Student Organization's use of space under their oversight. It also identifies the Shop Manager as the contact-person for EH&S and the Shop Safety Committee. This form must be adopted, then updated when operations / personnel / facilities change, or annually for any Student Organization. All updates must be copied to EH&S for every academic shop and/or maker space on SU property.

SHOP NAME: _____ APPROX. SQ. FT.: _____

BUILDING NAME: _____ ROOM #'s: _____

OUTDOOR SPACE? ☐ NO ☐ YES – WHERE / SIZE? _____

IS SHOP OPERATED BY A STUDENT ORGANIZATION? (Note: Faculty Advisor must complete annually.)

☐ NO ☐ YES – NAME/CONTACT OF FACULTY ADVISOR? _____

FACULTY ADVISOR EMAIL / PHONE: _____ / _____

ACCESS / USE OF THE SPACE IS GRANTED FOR HOW LONG? (Up to 1 year allowed.) _____

RISK ASSESSMENT OF HAZARDOUS PROCESSES / EQUIPMENT: (☒ Check AND circle all that apply):

☐ Risk assessment submitted previously (date) _____, and nothing's changed. (Go to page 2)

Consider all planned and possible work activities that may be conducted in the space based upon the following list of Shop / Makerspace work processes.

- ☐ **Metal working** including sheet metal forming, machining, grinding, riveting, cutting, threading, casting, forging, heat-treating, quenching, welding, brazing, soldering, 3D Printing, sputtering, etc...
- ☐ **Carpentry and woodworking** including cutting, drilling, sanding, carving, routing, grinding, planing, gluing, bonding, fastening, etc...
- ☐ **Surface modification and coating** including sandblasting, painting, surface preparation, laminating, burning, etching, masking, etc...
- ☐ **Glass work** including glass blowing, glazing, annealing, tempering, bonding, grinding, drilling, hot-work with glass materials, etc...
- ☐ **Electrical / electronic** work including equipment building, circuit design / building, wiring, control system building/repair, soldering, etc.
- ☐ **Plastics and Fabrics** work including machining, bending, burning, bonding, extruding, cutting, drilling, gluing, melting, forming, sewing, stitching, 3D Printing, etc...
- ☐ **Concrete / Rebar** work including form-building, rebar cutting, concrete mixing / pouring, destructive testing, etc...
- ☐ **Equipment Development / Model Building** work including model building, hydraulics building / use, compressed air use, equipment building, research equipment repairs / maintenance, etc...
- ☐ **Other Processes Not Listed:** _____

COMPLETED BY: _____ DATE: _____

EMAIL / PHONE CONTACT: _____ / _____

File: SHP-MNGR-Attachment 3- Shop-Space-Designation-form-MOU-FINAL-10-18-2019.docx

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Shop Safety Program Stanford University

Stanford University Shop Safety Program – Attachment 3

Academic Shop / Makerspace - Space Designation / MOU / Safety Contact Identification

MINOR SHOP PROCESS - EXEMPTION FROM THE SHOP SAFETY PROGRAM:

Many labs / student-use spaces have lower hazard shop processes (e.g. only hand tool, small hand-held power tools only, table-top 3-D plastic printer, etc.) that are ancillary to primary work activities of the space. For such locations, full enrollment in the Shop Safety Program is voluntary. Owner/managers of such spaces must still designate a “safety coordinator” for the Shop Safety Committee (SSC) to communicate with, but may opt-out of sending a representative from the work area to SSC meetings.

IS THIS SPACE EXEMPTED FROM THE SHOP SAFETY PROGRAM, BASED ON THE ABOVE CRITERIA?

☐ **NO** - THIS IS A SHOP / MAKERSPACE – (skip down to “FOR SPACES THAT ARE SHOPS” below)

☐ **YES** – NAME OF SAFETY COORDINATOR FOR SPACE: _____
SAFETY COORDINATOR EMAIL / PHONE: _____ / _____ [END]

FOR SPACES THAT ARE SHOPS / MAKERSPACES:

SHOP MANAGER NAME: _____ TITLE: _____

SHOP MANAGER CONTACT PHONE / EMAIL: _____ / _____

SHOP MANAGER IS: ☐ FACULTY ☐ STAFF ☐ STUDENT

MANAGER'S AUTHORITY DELEGATED BY: _____ TITLE: _____

TYPE OF SHOP: (Check all that apply)

- | | |
|--|--|
| ☐ Permanent Student Shop | ☐ Permanent Professional Shop |
| ☐ Temporary Shop / Student Org. Shop → Space Use Privileges Expire (date) _____ | |
| ☐ Temporary Field / Research Shop → Planned Use - End Date: _____ | |

RESTRICTED AREA DEMARCATED BY (Check all that apply):

- | | |
|---|--|
| ☐ Gate or Movable Barrier | ☐ Floor Paint / Floor Striping |
| ☐ Fence or Immovable Barrier | ☐ Personnel on watch restricting access |
| ☐ Orange cones / Temporary Barrier | ☐ Signage |
| ☐ Other (describe): _____ | |

SHOP ACCESS TO AUTHORIZED PERSONS ONLY - RESTRICTED BY (Check all that apply):

- | | |
|---|---|
| ☐ Locked Door(s) with key | ☐ Inside a larger controlled space |
| ☐ Electronic ID Access Control (24 hour) | ☐ Gate – With / Without latch |
| ☐ Personnel on watch restricting access | ☐ Signage |
| ☐ Other (describe): _____ | |

WILL THE SHOP MANAGER BE THE SHOP'S REPRESENTATIVE ON THE SHOP SAFETY COMMITTEE (SSC)?

☐ **YES** ☐ **NO** – NAME OF SSC REP FOR SHOP: _____

SSC REP EMAIL / PHONE: _____ / _____

COMPLETED BY: _____ DATE: _____

EMAIL / PHONE CONTACT: _____ / _____

File: SHP-MNGR-Attachment 3- Shop-Space-Designation-form-MOU-FINAL-10-18-2019.docx

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ATTACHMENT 4 – SOLO WORK – RISK REDUCTION PLAN



Stanford University – Academic Shop Safety Program Attachment 4 - Solo Work Risk Reduction Plan - Form

Instructions: Academic Shops on Stanford University property must comply with the following:

GENERAL SOLO WORK RULES IN ACADEMIC / STUDENT SHOPS

1. Shop access outside of supervised hours is restricted to specific "Authorized Persons" based upon the Shop Manager's discretion. Access for on-going solo work is not granted in any Academic Shop. On-going and/or routine solo work outside of 'normally scheduled operation and/or staffed hours' is strictly forbidden and may result in revocation of shop-use privileges by the Shop's owner department.
2. An "Authorized Person" may work solo in a Shop during 'normally scheduled operation and/or staffed hours' when communications means are made with the Shop Manager / Supervisor to assure the person's continuous safety and ability to communicate immediately with emergency responders.
3. When using large fixed powered shop equipment, flammable solvents, acids and/or etchants, heat and/or spark-producing equipment, at least one other Authorized Person must be in the shop area.

RISK MITIGATION PLAN FOR SPECIFIC SOLO WORK

When solo work must be conducted outside of "normally schedule operations and/or staffed hours", this Risk Mitigation Plan (Pages 3&4) must be completed and documented by the Solo Worker, in consultation with and approval by the Shop Manager, that address the following:

1. Requests to work alone after hours, or on weekends, must be approved by the Shop Manager case-by-case each time solo work is to be performed.
2. Consult with the Shop Manager when planning to work alone. What operations will be conducted, and what means for immediate communication in an emergency are needed during solo work?
3. Development of the Solo Work Risk Mitigation Plan must include review of the specific work activities, associated hazards and hazard controls to be used; and an emergency communication plan including contact name(s) and contact method(s).
4. If possible, solo workers should communicate with others in the building when working alone in the shop. Let them know when you arrive, what you're doing and when you leave.
5. Always avoid performing any high-hazard operations when working alone in the shop.
6. Never leave shop equipment unattended and operating.

Shop Managers must consider requests to work solo on a case-by-case basis. Any combination of the following must be considered to ensure control of the hazards of solo work to an effective degree:

- The authorized shop user's training and experience.
- Tasks and hazards involved in the planned solo work.
- The likelihood of an injury or illness that would prevent the shop user from calling for help.
- Consequences resulting from a worst-case scenario of sudden injury or illness.
- Deadlines, time-constraints and harried pressure to complete work that could negatively affect the shop user's safe work-methods and clear-headed work-approach.
- The shop user's physical condition and any constraints that may impact their safe work approach and methods. [Consult with local Human Resources Officer for guidance and compliance with the Americans with Disabilities Act (ADA)].
- The time of day solo work is to be conducted, for example, during normal business hours (7 am – 6 pm Monday through Friday) versus at night or weekends/holidays.
- The length of time the solo work is to be done (e.g. 5 minutes vs 5 hours vs 5 days)
- What Personal Protective Equipment should be worn during solo work?
- What Emergency Response Procedures, equipment and/or contact information is needed for solo work to take place? Besides the solo worker, who needs access to this information and equipment? How will they get this information prior to work commencing?
- Safe and unobstructed access to the nearest eyewash, shower, and fire extinguisher.



Shop Safety Program
Stanford University



Stanford University – Academic Shop Safety Program
Attachment 4 - Solo Work Risk Reduction Plan - Form

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Shop Safety Program Stanford University



Stanford University – Academic Shop Safety Program Attachment 4 - Solo Work Risk Reduction Plan - Form

Solo Worker: _____ Cell Phone: _____
Planned Work Activities: _____

Date / Time / Duration of Solo Work: _____ / _____ / _____

Location of Solo Work: _____

Solo Buddy Contact Name: _____ Cell Phone: _____

Emergency Contact Name: _____ Cell Phone: _____

Shop Manager Name: _____ Cell Phone: _____

RISK MITIGATION PLAN

The shop manager in conjunction with the solo worker may choose to implement any combination of the following risk-reduction options to control the hazards of solo work. These strategy options are to be implemented by honor system and mutually agreed to between Shop Manager and the Authorized Person on a case-by-case basis prior to the shop user being allowed to work solo in an academic shop:

HAZARD REDUCTIONS (Check all that apply)

- ☐ Always wear proper PPE including (list) _____
- ☐ Only hand-tool use is allowed when working solo.
- ☐ Limit higher-hazard solo work to normal week-day business hours when other personnel are likely to be nearby and able to respond to an emergency.
- ☐ Substitute low-hazard chemicals for high hazard chemicals while doing solo work. (e.g., use water-based paint instead of oil-based paint and solvents)
- ☐ Avoid setting up large-scale/hazardous projects when working solo.
- ☐ Adopt Shop Rules so that No High-Hazard Work is done solo nights and weekends, including:
 - ☐ Hazardous Chemical Use (Solvents, Etchants, Acids, Fuels, etc...)
 - ☐ Powered Machinery use
 - ☐ Hand-held Power Tools use
 - ☐ Heat / Spark / Arc / Open Flame / Smoke producing operations

ADMINISTRATIVE CONTROLS AND/OR "BUDDY SYSTEM" (Check all that apply)

- ☐ Set up a Buddy system – In person OR by cell phone / radio. Either way, your buddy must be able to recognize trouble and seek emergency help without the shop user's assistance.
- ☐ Develop and share specific emergency procedures with your buddy using this "Solo Work Risk Reduction Plan".
- ☐ Install an "emergency response" / "person down" app on your mobile device that will automate the call-out function in the event the solo worker stops moving or becomes horizontal for an extended period of time.
- ☐ Communicate with others in the building, building security, campus police or the shop manager. Let them know when you arrive, what you're doing and how long you intend to be there, and when you leave.
- ☐ Have your buddy check in on you every 10-15 minutes, OR....Check-in with your buddy every 10-15 minutes.
- ☐ Keep shop doors open (when there's no door window) or unlocked (with a door window) when solo work is happening.
- ☐ Setup your buddy in a safe place in the shop or shop-adjacent so they can be present and aware of your wellbeing if you must do high-hazard solo shop work.

[OVER]



Shop Safety Program Stanford University



Stanford University – Academic Shop Safety Program Attachment 4 - Solo Work Risk Reduction Plan - Form

DESCRIBE ANY OTHER PROCESSES, PLANS OR COMMUNICATION STRATEGIES TO ENSURE SAFE SOLO WORK FOR THIS LIMITED TIME:

SOLO WORKER SIGNATURE / DATE: _____ / _____

SHOP MANAGER SIGNATURE / DATE: _____ / _____

Instructions: Shop Manager, please keep completed copies of these forms (pages 3 & 4 only) on file for each time solo work is approved in your shop. Completed forms should be kept on file for 3 years after end of employment, or use of shop by Shop User.

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ATTACHMENT 5 – SHOP SAFETY COMMITTEE – MEETING AGENDA



Stanford University – Academic Shop Safety Program
Attachment 5: Shop Safety Committee Meeting Agenda Template

Stanford University - Shop Safety Committee AGENDA

[ENTER TIME AND DATE OF MEETING HERE]

[ENTER LOCATION OF MEETING HERE – BUILDING NAME AND ROOM NUMBER]

1. **Welcome** [SSC CHAIR PERSON]
2. **Review of last SSC Meeting Notes** [SSC SCRIBE / EH&S]
3. **Updates / Old Discussion Items** [SSC CHAIR PERSON]
 - a. [BULLET]
 - b. [LIST]
 - c. [DISCUSSION]
 - d. [ITEMS]
 - e. [HERE]
4. **Shop Safety – What's New OR Guest Speaker?** [EH&S SSC PROGRAM MNGR.]
5. **Review of any recent Shop WC Injuries, Near Misses and Lessons Learned** [EH&S SSC PROGRAM MNGR.]
6. **New Discussion Items** [SSC CHAIR PERSON]
 - a. [BULLET]
 - b. [LIST]
 - c. [DISCUSSION]
 - d. [ITEMS]
 - e. [HERE]
7. **Services Offered / Services Needed / Networking Opportunities between shops** [GROUP DISCUSSION]
8. **Determination of Date / Time / Location of next SSC meeting** [SSC CHAIR PERSON]
9. **Announcements** [GROUP DISCUSSION]

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ATTACHMENT 7 – ANNUAL SHOP SAFETY SELF ASSESSMENT

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

SHOP / MAKERSPACE SAFETY - PROGRAM ADMINISTRATION

1. Does your shop / makerspace have a copy of last year's self-assessment form and have all needed safety corrections documented on the form been completed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: If a self-assessment was completed last year, obtain a copy and determine current status of the shop. If necessary, contact EH&S [650-723-0448] for help.

2. Does your shop / makerspace have a current Shop / Makerspace Designation MOU form "Attachment 3" filed with the Shop Safety Committee and EH&S?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Every shop / makerspace must be designated for use for this purpose. To designate a space for use as a shop / makerspace, complete the Shop / Makerspace MOU Attachment 3 that identifies the space location, size, owner, shop manager, hazardous processes done in the space, and who will represent the shop / makerspace on the Shop Safety Committee. In addition, Student Groups using a shop / makerspace are granted space-use for up to one year using this MOU. Attachment 3 MOUs are available on the EH&S website. Contact EH&S [650-723-0448] with any questions.

3. Are "visitor spaces" versus access-restricted work areas clearly defined in the shop / makerspace? And, are only 'authorized' personnel, and their escorted visitors, allowed in restricted 'work areas' when wearing appropriate PPE?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Determine where visitors to the shop must stay, beyond which they must be escorted when entering into a shop work area / restricted space. Assure that visitors only enter shop work areas with an escort, and require that all people entering the restricted shop work areas wear appropriate PPE. Provide PPE to visitors as shop hazards may dictate.

Completion Date: _____

4. Does the shop / makerspace have a process to 'authorize' personnel to work unescorted in restricted work areas?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Determine what hazards are present in the shop and what locations are 'safe' for unescorted visitors to stay in. Determine what users must know about the shop / makerspace to 'work safe' and then implement training and other processes to "authorize" users to work unescorted, and to escort visitors if needed. Use "Attachment 1" of the Shop Safety Program to document each shop / makerspace user's authorization to work in the shop / makerspace. Use "Attachment 6B" for documenting specific training any shop / makerspace users undergo prior to being 'authorized' to work unescorted in the shop / makerspace.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

5. Does the shop / makerspace have a policy to prohibit solo work, or develop a "Solo Work Risk Reduction Plan" using Attachment 4 when users must do solo work?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: The Shop Safety Committee has adopted a policy that Solo Work is hazardous and must not be routinely done. If solo work must be done, a "Solo Work Risk Mitigation Plan" must be completed before solo work is done. This plan is completed by the solo worker and the shop manager using Attachment 4.

Completion Date: _____

6. Are shop / makerspace users and visitors provided with eye, face, and hand protection such as safety glasses, face shields and gloves where needed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure shop users and visitors have appropriate safety equipment and PPE available for use at no cost to them. Contact EH&S [650-723-0448] with any questions.

7. Are shop / makerspace users and visitors provided hearing protection, such as ear plugs or ear muffs, while operating or when nearby noisy pieces of equipment? Are locations where hearing protection is required clearly marked with signage?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Workers exposed to excessive noise may need to be enrolled in the hearing conservation program. More information about this can be found on the EH&S website, or by calling EH&S at [650-723-0448].

SHOP / MAKERSPACE - SAFETY INFORMATION / TRAINING

8. Does the shop / makerspace have Job Safety Analyses, Standard Operating Procedures, or Operations Manuals that address hazards for all power tools and hazardous processes?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Obtain operator's manuals from the manufacturer or write and/or adapt an existing job safety analysis for each power tool, piece of equipment or shop process. Ensure that the JSAs are kept in the "Shop Safety Reference Manual" and that shop users are familiar with the manual. Many JSAs are available online, can be written by shop users or can be adapted from other shop's JSAs. Contact EHS at [650-723-0448] for help in developing JSAs and building your Shop Safety Manual.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

9. Is there a file or binder of Safety Data Sheets (SDS) for each chemical stored or used in the shop? Or, are all shop personnel trained on how to access and use the online campus SDS access system?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Information regarding SDSs is available at EH&S Chemical Safety Database [<https://ehs.stanford.edu/services/safety-data-sheet-database>]. Obtain missing SDSs from product vendors and populate a hard-copy of the SDS in the Shop Safety Manual. Or, post clear information on how to access the campus' online SDS system, and train all shop users on how to access/use the online system. Contact EH&S at [650-723-0448] for further information / support as needed.

Completion Date: _____

HOUSEKEEPING

10. Do all work areas have adequate ventilation, particularly for hazardous operations (e.g., welding, soldering, painting, spray coating, solvent use)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact EH&S [650-723-0448] to have ventilation evaluated if there is a concern. Provide additional exhaust ventilation for operations that create dust, fumes, mists, or vapors.

Completion Date: _____

11. Are exit/egress pathways maintained clear and all exits clearly marked?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Clearly define egress pathways and enforce keeping these pathways clear at all times. Assure that exit signs are visible from all shop locations.

Completion Date: _____

12. Are shop / makerspace users required to clean up their work area(s) prior to departure?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: All shop / makerspace users are required to clean up after themselves. It must be part of shop / makerspace culture that every user is responsible for the mess they make, and clean up to return the work area to a "work ready" condition for the next user.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

FIRE / EMERGENCY SAFETY

13. Are egress pathways / exit aisles clearly identified and kept unobstructed to exit doorways?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all exit pathways are at least 32" wide and kept clear. Remove any obstructions and stabilize or secure anything that could fall into an egress pathway during an earthquake.

Completion Date: _____

14. Are emergency evacuation routes clearly identified using a building map, and communicated to shop / makerspace users as part of their training?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: All shop / makerspace users must know the evacuation plan for the space, where to go for a muster-point during an emergency and how to get there. This information must be part of any shop / makerspace user's orientation / training before using the space.

Completion Date: _____

15. Are fire alarm pull boxes and fire extinguishers clearly identified and unobstructed?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that all fire alarm pull boxes and fire extinguishers are clearly labeled. Remove any obstructions in front of fire alarm pull boxes and fire extinguishers.

Completion Date: _____

16. Are portable fire extinguishers tagged with current annual inspections?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact EH&S to have fire extinguishers inspected if they have not been inspected within the last 12 months. Every extinguisher should have a tag attached to it with the expiration date indicated. The extinguisher is OK if the date is more than 30 days in the future.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

17. Are the emergency eyewashes tested (flushed) monthly, with the tests documented?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Assign a shop user to test all emergency eyewashes monthly. Document each test in a logbook or on an attached tag. Contact LBRE for repairs. Call EH&S for a tag [650-723-0448], if one is needed.

Completion Date: _____

18. Are all eyewash and emergency shower stations free of obstructions that would prevent quick access by someone temporarily blinded?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Remove all obstructions from emergency eyewashes and showers and the pathways leading to them. Keep access open at all times.

Completion Date: _____

19. Are flammable gas cylinders and oxygen cylinders separated by a fire-rated wall or stored at least 20 feet apart?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Separate flammable gases from oxygen by a fire wall or a distance greater than 20 feet. Note: Storage of oxygen and acetylene cylinders on welding carts that are designed for the purpose/use of flame-welding / cutting are permitted.

Completion Date: _____

20. Are covered metal containers used for oily, solvent, paint-soaked waste?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that covered metal containers are provided and used for the disposal of oily and paint-soaked rags.

Completion Date: _____

ELECTRICAL SAFETY and ENERGY ISOLATION

21. Are electrical panels accessible, with at least 36" clearance at the front of the panel?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure that there is a minimum of 36" of clearance in front of all electrical panels, and a 32" wide pathway to access the panel area.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

22. Is there a procedure for removing or isolating dangerous (damaged, broken, unguarded) power-tools or equipment from service?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Establish a procedure such as using "out of service" tags to identify and prevent the use of dangerous items.

Completion Date: _____

23. Does your shop use Energy Isolation - Lock out Tag out (EI-LOTO) to isolate equipment before machine guards are removed, or before damaged equipment can be repaired?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable – we don't do this work)

Corrective Action: If your shop builds, repairs, modifies or otherwise works with or on equipment that is 'hard wired' or 'hard plumbed' to a facility, utility or other energy source, EI-LOTO must be practiced. Contact EH&S Occupational Safety and Health Team for more information on Stanford University's Energy Isolation - Lock out Tag out (EI-LOTO) Program.

Completion Date: _____

HAND TOOLS and POWERED HAND TOOLS

24. Are all hand-operated tools free of defects (such as cracked handles, frayed or damaged cords, missing parts, missing guards, etc.) that make them unsafe?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Remove defective tools from service. Either have them repaired, or replace them with new equipment that is in good working condition.

Completion Date: _____

POWERED MACHINERY

25. Do all powered machines have manufacturer-supplied, or otherwise appropriate, machine guards on all pinch points, crush points, rotating parts, blades, tooling, chucks, etc.?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Install and use manufacturer-supplied machine guards on all powered equipment. For equipment that is too old for manufacturer-supplied guards, assure that all areas of the equipment have a 'home-made' guard to prevent accidental access to powered parts. Contact EH&S [650-723-0448] for assistance with machine guarding assessment, design and use.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

26. Does the shop have a process to change out and / or sharpen tooling and drill bits when they become dull or damaged?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Sharp tooling allows powered equipment to perform safely and predictably, run without excessive heat buildup and machine workable materials more quickly. Shops should have a process to change out dull or broken tooling routinely.

Completion Date: _____

27. Does the shop require users to remove hand/wrist jewelry, neck and other body jewelry, loose fitting clothing, as well as require that long hair is tied or worn under a hat or cap while operating powered equipment?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: All shop users must prepare to safely work with powered equipment by removing jewelry from hands / wrists, loose neck and other body areas, when working with powered equipment. Hair and loose-fitting clothing must always be under control and kept away from powered equipment.

Completion Date: _____

MATERIAL HANDLING / CRANES / HOISTS / RIGGING

28. Are materials in excess of 50 lbs routinely handled with the assistance of material handling devices such as pallet jacks, pump trucks?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Ensure proper lifting techniques and equipment are used. For questions about safer material handling devices, contact EH&S at [650-723-0448].

Completion Date: _____

29. If materials in excess of 50 lbs are handled with the assistance of devices such as cranes, chain-falls, hoists, rigging, etc., has this equipment has been load tested / inspected by an OSHA certifying inspection agent?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Cal-OSHA requires cranes, hoists, slings, hooks and other lifting gear be inspected and load tested. EH&S can arrange for and provide training for safe operation / load testing / rigging inspections. For questions about cranes, hoists or rigging devices, contact EH&S at [650-723-0448].

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

30. If materials in excess of 50 lbs are handled with the assistance of forklifts or other type of industrial lift-trucks, are the lift-trucks routinely inspected for safety before use and are operators 'licensed' to operate these truck(s) through the EH&S Industrial Lift Truck Safety / Training program?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Cal-OSHA requires forklifts, pallet stackers and other industrial lift-truck operators are trained and licensed prior to equipment use. In addition, equipment must be routinely inspected for safety through documented safety inspections. EH&S provides classroom and hands-on training for safe equipment inspection, operation and licensing for using Industrial Lift-Trucks. For questions about Industrial Lift Truck Safety, or to enroll shop users for training / licensing, contact EH&S at [650-723-0448].

Completion Date: _____

HAZARDOUS MATERIALS and HAZARDOUS COMMUNICATIONS (HAZCOM)

31. Are emergency procedures and telephone numbers clearly posted?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Post emergency procedures, evacuation routes and muster-locations, telephone numbers (police, fire, ambulance) near the telephone(s) where shop users have access.

Completion Date: _____

32. Is a Shop Hazard Communication Program HAZCOM template completely filled out and posted in the shop? Does a 'shop safety orientation' for personnel include reference to the HAZCOM information for the shop?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Obtain and complete the HAZCOM template available from EH&S [650-723-0448]. Post in a conspicuous place, and include training about HAZCOM for the shop as part of your shop's user safety training.

Completion Date: _____

33. Are all containers of chemicals labeled with the contents and primary hazard(s)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Properly label all chemical containers, including stored and temporary containers.

Completion Date: _____



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 7 Annual Shop Safety Self-Assessment

34. Are incompatible chemicals stored separately (e.g., acids separate from bases, oxidizers separate from flammables)?

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact EH&S [650-723-0448] for assistance with chemical storage and segregation, or go to EH&S website for further information on chemical storage safety.

Completion Date: _____

35. Are all compressed gas cylinders adequately secured with non-combustible restraints to keep the cylinder(s) from falling during an earthquake? (If chains are used, two should be present for each cylinder.)

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Contact EH&S [650-723-0448] for advice on how to properly secure cylinders. Bench clamps may not be used for securing cylinders. Train shop personnel to cap compressed gas cylinders when not in use.

Completion Date: _____

36. Has a current chemical inventory materials door sign (provided by EH&S) been posted at the shop / makerspace entrance following your most recent chemical inventory update (within last 12 months?)

☐ Yes (Satisfactory) ☐ No (Needs Correction) ☐ N/A (Not Applicable)

Corrective Action: Locate and post the updated door sign at the shop / makerspace entrance or contact EH&S at [650-723-0448] for another copy.

Completion Date: _____

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ATTACHMENT 8 – SHOP / MAKERSPACE – SET UP / MANAGEMENT GUIDE

Stanford University – Shop Safety Program – Attachment 8 Shop / Makerspace Set-up and Management Guide

Attachment 8: Shop / Maker Space Set-up and Management Guide

Instructions for Shop / Maker Space Manager:

This is a Shop / Maker Space Set-up and Management Guide for any person delegated the responsibility to set up and manage a Shop and/or Maker Space on Stanford University property. By following this guide, a Shop / Maker Space Manager will put into place all necessary hazard identification, control and user-training requirements for integrating the Stanford University Shop Safety Program into their shop.

Instructions for ShLab Safety Coordinator:

A “ShLab” is a space where shop equipment or work-processes are used as part of laboratory operations. As a ShLab Safety Coordinator, you are responsible to act as a liaison between your lab and the Shop Safety Committee, and for implementing the Shop Safety Program as may be appropriate for your lab. Familiarize yourself with this document and the Shop Safety Program on the EH&S website. You are performing the “shop manager” role for your lab.

Shop processes and equipment-use in labs are generally ancillary to lab operations. Therefore, participation in the Shop Safety Program may be voluntary as described in the “Lab Exemption” in the “Applicability and Scope” section of the Shop Safety Program. However, EH&S must have your contact information to keep you and your lab informed of shop safety program requirements and updates as they may impact your lab’s safety and operations. Complete Attachment 3 MOU of the Shop Safety Program and send it to the EH&S Shop Safety Program Manager at this email. [<shopsafety-ehs@stanford.edu>](mailto:shopsafety-ehs@stanford.edu)

Building the Physical Shop / Maker Space

Every Shop / Maker Space must have certain physical space / access characteristics to ensure safety can be maintained by the manager of the space. When setting up a new shop / maker space, or modernizing / repurposing an existing space that will contain any of the work-processes outlined in the “Applicability” section of the Shop Safety Program, the Shop / Maker Space Manager should incorporate the following into the planning and installation of the Shop / Maker Space:

1. Determine the location and boundary of the “Restricted Area” within the shop / makerspace where high-hazard work will take place. Clearly mark the “Restricted Area” using signage, floor-paint-striping, and/or other means to create visual cues of the boundary.
2. Develop and install a location where “Authorized Users” and “Visitors” may enter the shop / maker space, remove and safely store their “street clothing and gear” and put on PPE prior to entering the “Restricted Area” of the shop / makerspace.
3. Ensure access control of the “Restricted Area” of the space by installing physical barriers to non-authorized users such as doors that lock and require a key, combination-lock or other “user-knowledge” to access the space and conduct high-hazard work activities and use high-hazard equipment.
4. Document the formation and location of the Shop / Maker Space, and inform the Shop Safety Committee and EH&S of the name, title and contact information for the person responsible for shop safety for the space using [Attachment 3 MOU](#).



**Stanford University – Shop Safety Program – Attachment 8
Shop / Makerspace Set-up and Management Guide**

Qualifying and Authorizing Shop / Makerspace Users

1. Any person who works with shop equipment and processes must be “authorized” by their PI/Instructor/Supervisor, as well as the “Shop Manager” prior to working unescorted in a shop. If the “Supervisor” is the same person as the “Shop Manager”, then the Shop Manager completes both halves of Attachment 1 for each “Authorized Person” who may work in their shop. This is a documented process using [Attachment 1](#).
2. All “Authorized Persons” must be trained on, and remain current to, each shop / makerspace’s specific safe-work requirements including but not limited to general safe work rules, personal protective equipment use, and solo work. Training on these Stanford University Shop Safety Requirements are documented by each Shop Manager using [Attachment 2](#) as well as any other training documentation a Shop Manager may deem necessary to ensure safety within their shop operations. Use [Attachment 6B](#) Training Roster to document any “shop specific” safety training given to shop users.
3. “Authorized Persons” may only work on or with shop equipment and conduct shop processes that they have been ‘Authorized’ by the Shop Manager to do.
4. Once authorized, “Authorized Persons” may enter and work unescorted in a shop’s “Restricted Area”.

Managing Shop “Visitors”

Visitors:

1. May not enter the “Restricted Space” of any shop unless escorted continuously by an “Authorized Person” in that specific shop.
2. May be escorted within the shop’s “Restricted Area” as long as the visitors are wearing the same PPE as required by all shop users.
3. Must remain within 10 feet or less of the Authorized Person escorting them.
4. Must remain closer as needed to ensure visual and audio contact with the Authorized Person at all times while in the “Restricted Area” of a shop.

Authorized Persons:

5. May escort visitors within the shop’s “Restricted Area”, but only with the Shop Manager’s permission.
6. Must provide PPE to all visitors as shop rules dictate, and ensure all visitors being escorted are properly wearing their PPE at all times within the Restricted Area of the shop.



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 8 Shop / Makerspace Set-up and Management Guide

Managing Contractors using the Shop

Contractors and Joint Projects in Shops

When using their own equipment and shop processes, when not using Stanford shops or Stanford shop equipment, and when not working in conjunction with or around Stanford University personnel, Contractors must:

1. Follow their own Shop Safety Program and procedures.
2. Provide the University project manager and/or Shop Manager evidence of their Shop Safety Program upon request when working on Stanford-owned property.
3. Provide all their own shop equipment, materials and supplies unless negotiated by contract with the University the use of University facilities and equipment.

Contractors using Stanford University-owned Shops and Shop Equipment

The Shop Manager must:

1. Use the same authorization / orientation process used for Stanford University Shop Users by completing [Attachment 1](#) and [Attachment 2](#) for each contractor-employee using a Stanford-owned and managed shop. The Contractor Representative hiring and overseeing the work conducted by contractor personnel must act as the "Supervisor" and document contractor authorization and orientation / training to the Stanford University Shop Safety Program prior to allowing the contractor-personnel to work in any Stanford University owned / managed shop.
2. Approve and authorize each contractor-employee to work in their shop prior to allowing Contractor-use of the shop by signing the bottom half of [Attachment 1](#) and [Attachment 2](#) for each authorized contractor-employee.

Joint Projects – When University Personnel and Contractor(s) work in a University-owned shop concurrently:

The Shop Manager is the controlling person in the shop / makerspace. Only the Shop Manager may authorize someone to work in the shop / makerspace. Contractors must follow "Joint Project" requirements as outlined below when working in the area of or with Stanford University shop / makerspace users.

The Shop Manager must:

1. Hold joint meetings with all Stanford and Contractor personnel in attendance who will be working in the area to promote understanding of planned work, safe work practices and open lines of communication amongst workers regardless of employer.
2. Ensure Contractors using University equipment and property follow University safe-work procedures and review JSAs and SOPs developed by the Stanford University shop as part of their training/orientation to the shop.
3. Document Joint Project / Contractor orientation/training by using a training roster ([Attachment 6B](#)) and keep completed training records for length of the Contractor project plus 3 years.



**Stanford University – Shop Safety Program – Attachment 8
Shop / Makerspace Set-up and Management Guide**

Shop Manager – Ongoing Management Requirements

All Shop / Makerspace Managers must:

1. Ensure the physical space of the shop / makerspace complies with the guidelines outlined above.
2. Ensure only "Authorized Persons" conduct work in the "Restricted Area" of their shop. Documents this process by completing the bottom-half of [Attachment 1](#) for each authorized shop user.
3. Ensure every "Authorized Person" in the shop has read, been given, and signed off receiving a copy of the Stanford University Shop Safety Requirements using [Attachment 2](#).
4. Add-to their specific shop / makerspace's "Authorized Person Safety-training" any additional hazard control measures (in addition to Attachment 2) such as SOPs, JSAs, specific equipment / processes orientation, etc. that the Shop / Maker Space Manager requires to ensure safe work within their area. Document this additional training using [Attachment 6B](#).
5. Ensure any Contractor-employee working in any Stanford-owned shop / maker space goes through the same "Authorized Person" safety orientation as other shop users, and have their training documented by completing, at minimum, [Attachment 1](#) and [Attachment 2](#).
6. Conduct "Joint Projects" meetings when contractor work is present as well as other responsibilities outlined in the "Contractors and Joint Projects" section above. Document this additional training using [Attachment 6B](#).
7. Purchase and keep in stock Personal Protective Equipment (PPE) for use by Authorized Personnel and Shop Visitors based upon a hazard assessment / control measures of their equipment and processes.(may use [Attachment 7](#) or similar)
8. Maintain training records of all Shop "Authorized Persons" working in the shop using a training roster ([Attachment 6B](#) or similar). Training records are kept for duration of shop-work plus 3 years.
9. Provides refresher training to "Authorized Persons" on their shop's safety program, whenever new hazards and control measures are identified, by using [Attachment 2](#) and any other shop safety training as the Shop Manager may require.
10. Regularly attends Shop Safety Committee (SSC) meetings as a member, or designates a Shop / Makerspace representative to be a member, of the SSC.
11. Oversee conducting an annual (at minimum) shop safety self-assessment / safety inspection ([Attachment 7](#)). May implement more frequent inspections to maintain shop safety within their shop as they deem appropriate.
12. Assure a hazard assessment / control review is conducted for any new shop equipment / processes.
13. Conduct new training of Authorized Persons as may be needed to ensure safety with new equipment or new shop processes.



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 8 Shop / Makerspace Set-up and Management Guide

Training Requirements

Shop “Authorized Person” / Shop User

Must be trained on:

1. Stanford University's Shop Safety Program by, at minimum, reading / signing [Attachment 1](#) and [Attachment 2](#), and
2. Any other training required to orient a Shop User to a specific shop's safe work processes and practices, and
3. How to escort any “Visitors” to a shop they have been authorized to work in.

Shop / Maker Space Manager

1. Must know and implement their [roles and responsibilities](#) in the application of this program by following all requirements of this guideline.
2. Be trained on Shop Management as may be appropriate to ensure safety in location(s) and operation(s) under their purview (consult with EH&S for questions / help),
3. Assures proper documented training of people reporting to them and other “Authorized Persons” working in the shop, and
4. Assures compliance with the Shop Safety Program by all who fall under their oversight.

ShLab Safety Coordinator

Is trained on:

1. The contents of the Shop Safety Program and how it may apply to shop work and equipment used in their lab.
2. Their responsibilities to train other members of their lab in the specific shop safe work processes in their lab as may be appropriate.
3. The Shop Safety Program “[Record-keeping](#)” responsibilities, management of the annual shop safety self-assessment program in their lab, and other duties as outlined in the “[Roles and Responsibilities](#)” section of the Shop Safety Program.
4. Conducting an annual (at minimum) shop safety self-assessment / safety inspection for the lab using [Attachment 7](#).

Contractors and Joint Projects in Shops

Contractors follow their own Shop Safety Program and procedures and must train their personnel when using their own equipment and shop processes. The Contractor provides the University evidence of their Shop Safety Program upon request. The Contractor provides their own shop equipment and materials unless negotiated by contract with the University for use of University equipment and materials.

For Joint Projects where University personnel and Contractor(s) are working in the same University-owned shop at the same time, the University's Shop Manager is in control of the shop and must hold joint “tailgate” meetings with all personnel in attendance who will be working in the area to promote understanding of safe work practices, and open lines of communication amongst Shop Users. Contractors using University equipment and property follow University safe-work procedures and review and follow JSAs and SOPs as developed by the University's



Shop Safety Program Stanford University

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Shop Manager as part of their training. Contractor training conducted by Stanford University is documented by the University's Shop Manager using a training roster [Attachment 6B](#).

Record Keeping Requirements

Shop / Maker Space Manager keeps the following documentation:

1. Completes bottom half of [Attachment 1](#) for each person who must become an "Authorized Person" to work in a shop. If the PI / Instructor / Supervisor is also the Shop Manager, completes both top and bottom halves of [Attachment 1](#) for each authorized shop user. This must be kept for duration of employment, tenure or work-study term plus 3 years.
2. For Student Shops, use [Attachment 1](#) for a duration of Authorization for shop use for a semester up to a maximum of 1 year. At the end of 1 year, the Student may be re-authorized for ongoing shop use of up to another year if their academic program dictates need. This must be kept for duration of academic program plus 3 years.
3. A signed copy of page 3 of the "Shop Safety Policy" ([Attachment 2](#)) given to each and every "Authorized Person" working in their shop. This must be kept for duration of employment, tenure or work-study term, or academic program duration, plus 3 years.
4. Copies of any training rosters ([Attachment 6B](#)) and training content. Keep for 3 years.
5. Copies of any procedures / protocols developed for shop safety. Keep for as long as they are current.

ShLab Safety Coordinators

Must keep the following documentation:

1. Authorized Person(s) who conduct shop-processes in labs documents ([Attachment 1](#) and [Attachment 2](#)) for the duration of their lab work plus 3 years.
2. Any training rosters ([Attachment 6B](#)) for 3 years.
3. Any Safe-work Protocols / SOPs / Other Procedures as long as they are current.
4. Any JSAs that are developed by the Department to address Shop Safety concerns for as long as they are current.
5. "Shop Self Assessment" forms ([Attachment 7](#)) completed for the lab for the previous two years.



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 8 Shop / Makerspace Set-up and Management Guide

Resources

EH&S Programs and Support

[LINKS TO EH&S PROGRAMS AND WEBSITE INFO TO BE INSERTED HERE]

California Title 8 – Cal-OSHA General Industry Code of Regulations

Compressed Gas and Air Equipment	http://www.dir.ca.gov/Title8/sb7g9.html http://www.dir.ca.gov/Title8/3301.html http://www.dir.ca.gov/Title8/4650.html
Cranes and Hoisting Equipment	http://www.dir.ca.gov/Title8/sb7g13.html
Ergonomics	http://www.dir.ca.gov/Title8/5110.html
Fire Safety	http://www.dir.ca.gov/Title8/sb7g27.html
Flammable Liquids, Gases and Vapors	http://www.dir.ca.gov/Title8/sb7g20.html
Hand and Portable Power Tools	http://www.dir.ca.gov/Title8/sb7g3a20.html
Hazardous Energy – Lock out Tag out	http://www.dir.ca.gov/Title8/3314.html http://www.dir.ca.gov/dosh/etools/08-003/LockoutSteps.pdf http://www.dir.ca.gov/dosh/etools/08-003/P08-00302.pdf
Hazardous Materials	http://www.dir.ca.gov/Title8/sb7g16.html
Housekeeping / Work Area Conditions	http://www.dir.ca.gov/Title8/sb7g1a4.html http://www.dir.ca.gov/Title8/sb7g1.html
Machine Guarding	http://www.dir.ca.gov/Title8/sb7g8.html http://www.dir.ca.gov/Title8/sb7g6a37.html http://www.dir.ca.gov/Title8/sb7g6.html
Noise	http://www.dir.ca.gov/Title8/sb7g15a105.html
Personal Protection (PPE) / Safeguards	http://www.dir.ca.gov/Title8/sb7g2a10.html
Safe Work Practices	http://www.dir.ca.gov/Title8/sb7g2a7.html
Vehicles and Shop Mobile Equipment	http://www.dir.ca.gov/Title8/sb7g4.html
Welding / Cutting Operations – Electric	http://www.dir.ca.gov/Title8/sb7g11.html
Welding / Cutting Operations - Gas	http://www.dir.ca.gov/Title8/sb7g10.html

Federal OSHA:

Machine Controls Interlocks and Fail-safe Design Guideline document:

http://www.osha.gov/OshDoc/Directive_pdf/CPL_02-00-147.pdf



Shop Safety Program Stanford University

Stanford University – Shop Safety Program – Attachment 8 Shop / Makerspace Set-up and Management Guide

Shop / Makerspace Safety Program - Implementation Forms

(For use by Shop Manager / ShLab Safety Coordinator as may be appropriate)

Available on EH&S Shop Safety Program website.

- Attachment 1: "Authorized Person" qualification for Shop Work – Template
- Attachment 2: Shop Safe Work Requirements, Practices and Procedures - Template
- Attachment 3: Academic Shop Space Designation / MOU / Identification - Form
- Attachment 4: Solo Work Risk Reduction Plan
- Attachment 6B: Shop Safety Training Roster – Template Form
- Attachment 7: Shop Safety Self-assessment Template
- Attachment 8: Shop / Maker Space Set-up and Management Guide

File: SHP-MNGR-Attachment-8-Shop-Makerspace-Setup-MngmtGuideline-FINAL-10-18-2019.docx

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